

PREPARATORY PROBLEMS



CUBAN NATIONAL CHEMISTRY CONTEST019



Ministerio de Educación
República de Cuba

PREPARATORY PROBLEMAS

National Chemistry Contest 2019

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Preface

To the olympists

This booklet is addressed to Cuban students and mentors who will take part in the National Chemistry Contest in 2019, high school level, course 2018-2019.

The Preparatory Problems that are here presented were carefully selected to exercise the *Topics of advanced difficulty* that will be evaluated in the next edition of the olympiad. Hence, this material should be used as a reference for the training after the provincial contest. The proposed problems and the answers will be debated in the National Seminar for Mentors that will be held in Havana on October 8 to 12, 2018. In this event provincial and national mentors will share thoughts with the members of the scientific committee that will elaborate the contest.

Similarly as in the previous edition, the results of Cuban students in the National Chemistry Contest from 2018 and In the international chemistry olympiads form 2017 and 2018 are included. Besides, the main organic reactions that should be known by 12th graders are presented in a novel organization, different and moderner than in the current contest program.

This year, the national exam will be elaborated by teachers from the Faculty of Chemistry of the University of Havana. It will be qualified by provincial and national mentors. With this purpose, two independent tribunals will be created. The members of the tribunal and the results of the olympiads will be informed to the competent authorities in due time.

To all the students and mentors, I sincerely wish you a lot of success in the coming national and international events!

The author

Havana, September 2018

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant

$$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$$

Equation of ideal gas

$$p \cdot V = n \cdot R \cdot T$$

Gases constant

$$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

First Principle of Thermodynamics

$$Q = \Delta E + W$$

Zero on Celsius scale

$$273,15 \text{ K}$$

Gibbs' free energy

$$\Delta G = \Delta H - T \cdot \Delta S$$

Faraday's constant

$$F = 96485 \text{ C} \cdot \text{mol}^{-1}$$

Nernst's equation

$$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$$

Ionic product of water at 25 °C

$$K_W = 1,00 \cdot 10^{-14}$$

Arrhenius' equation

$$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$$

Standard conditions

$$p^\circ = 100 \text{ kPa}$$

$$c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$$

$$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$$

$$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$$

$$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$$

PERIODIC TABLE OF ELEMENTS

1 1A																18 8A	
1 H 1.008	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)
		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0		
		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)		

PROBLEMS

Topics of advanced difficulty

Common exam

Stoichiometry: Calculation of transformation relations, sample composition and empirical formulas of organic and inorganic compounds.

10th grade

Inorganic: Chemical properties of the of representative elements and most important transition elements, including the identification of substances/ions.

Structure: Representation of molecules and ions and description of the covalent bond by Lewis' formulas, MOT, VSEPR, resonance theory and hybridization.

Note: Students must be familiarized with the nomenclature and the chemical notation of simple organic substances.

11th grade

Equilibrium: Relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications to gas phase equilibrium, dissolution-precipitation and in redox processes. Factors influencing the dissolution-precipitation of ionic and molecular substances in water. Calculation of the solubility in complex systems.

Note: The students must be able to linearize equations, construct the corresponding graphs and interpret them.

12th grade

Equilibrium: Relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications to redox processes.

Organic: Aromaticity. Aromatics substances: method of obtention and chemical properties.

Problem 1. An unknown element¹

When heating a sample containing a simple substance **A** and fluorine at 900 °C three compounds are formed. The compounds (**B**, **C** y **D**) are crystalline solids with melting points below 150 °C. Fluorine contained in **C** represents 36,7% mass of its composition and fluorine contained in **D** represents 46,5% mass of its composition. When **B** is treated with HOSO_2F at -75 °C compounds **E** and HF are produced. **E** possesses two fluorine atoms in its formula and the reaction proceeds stoichiometrically (mol to mol).

When 0,450 g of **C** react with excess of mercury 53,25 mL of gas **A** are released and mercury difluoride is formed. Temperature is 25,0 °C and pressure is 101,0 kPa.

- 1.1. Calculate the relative atomic mass of **A**.
- 1.2. Identify the formulas of **A**, **B**, **C** and **D**. Show your calculations.
- 1.3. Represent the electronic geometry of **B** and **C** (spatial formula) using the Valence Shell Electron Pair Repulsion model (Gillespie's model).

The mixture of **B**, **C** and **D** was reacted with water. **C** with water forms **A**, oxygen and aqueous solutions of AO_3 and HF. It is known that each mole of **C** gives $\frac{2}{3}$ mol of **A** and $\frac{1}{3}$ mol of AO_3 . **B** reacts with water forming **A**, oxygen and hydrofluoric acid. **D** with water gives AO_3 and HF in aqueous state.

- 1.4. Write the equation of the three reactions.

The quantitative reaction between water and a mixture of **B**, **C** and **D** formed 60,2 mL of gases (measured at 290 K and 100 kPa). The oxygen in the mixture represents 40,0% (in volume). The amount of AO_3 dissolved in water was reacted with $3,6 \cdot 10^{-3}$ mol of aqueous FeSO_4 and sulfuric acid, giving iron(III) sulfate, gaseous **A** and water.

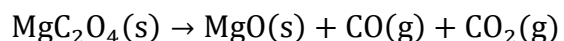
- 1.5. Calculate the molar composition (in per cent) of the mixture of **B**, **C** and **D** that was reacted with water.

¹ National Chemistry Contest. Cuba, 2006. Day 2. 10th grade. Problem 2.

Problem 2. Thermogravimetry²

Thermogravimetry is an analytic method to determine the composition of solids that decompose under heating. The mass change, which is measured during the heating process, provides information about the composition of the analyzed substance.

A mixture of calcium oxalate and magnesium oxalate is heated up to 900 °C. During the process the mass of the mixture is constantly measured. It is known that there are two decomposition reactions around 400 °C:



2.1. When the temperature reaches 700 °C a third decomposition is observed. Write the equation corresponding to this process.

2.2. Calculate the composition (in mass percent) of the previous mixture, if at 500 °C the mass was 3,06 g and at 900 °C the mass was 2,03 g.

The gravimetric method was tested. The molar mass of the carbon was calculated and compared with the reference data. The researcher heated 7,30 g of calcium oxalate and got the results:

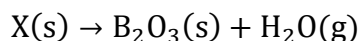
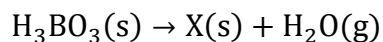
Temperature (°C)	90	250	500	900
Mass (g)	7,30	6,40	5,00	2,80

2.3. ¿Which is the cause of the first decrease in the mass?

2.4. Calculate the molar mass of carbon using the data that appears in the table.

Sometimes, the gravimetric method allows the discovery of new substances. For example, gravimetry of boric acid reveals the existence of compound **X**. Heating boric acid in two steps is accompanied with a reduction of the solid mass. Above 100 °C compound **X** experiences a decomposition.

Experimental results:



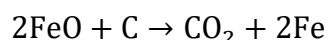
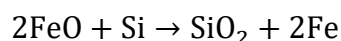
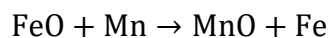
Temperature (°C)	40	100	250
Masa (g)	6,2	4,4	3,5

2.5. Calculate the empirical formula of **X**.

² Natural Chemistry Contest. Cuba, 2007. Day 1. Common exam. Problem 4.

Problem 3. Steel production³

Melted iron from high furnaces is a mixture of iron (approximately 94% mass), carbon (approximately 3% mass) and the rest is silicon and manganese, mainly. This is one of the main raw materials for the production of steel (allow of carbon and iron with 0,2% mass of carbon). In the process of steel manufacture, approximately half of the iron in melted iron is oxidized by the air to iron(II) oxide; then, after melting, this oxide reacts with the other components present in the following equations:



The rest of unreduced iron(II) oxide is treated with a mixture containing 90% of Mn, 6% of C and 1% of Si, denominated ferromanganese. Oxides MnO and SiO₂ is transformed in other compounds that when melted together with iron float on the surface because are less dense and can be separated. Assuming that the mole amount of silicon is 3/2 of the mole amount of magnesium in melted iron.

- 3.1. Calculate the mass percent of manganese and silicon in melted iron.
- 3.2. Calculate the ferromanganese mass that must be used to obtain a steel with the characteristics mentioned. Suppose that one ton of melted iron is used.

Problem 4. Formulas of mineral⁴

It is a common practice to express the formula of the minerals as a mixture of oxides. For example, the formula of kaolin, the main constituent of clay, may be written to Al₂O₃·2SiO₂·2H₂O. In this problem your task is to find the formula (oxide composition) of several important minerals.

Beryl is the most important mineral containing beryllium, it is a tertiary oxide of beryllium, silicon and aluminum. In beryl the mass ratio Be:Al is approximately 1:2 and the molar ratio Si:O is 1:3.

- 4.1. Calculate the empirical formula of beryl.

³ Natural Chemistry Contest. Cuba, 2008. Day 2. Problem 1.

⁴ National Chemistry Contest. Cuba, 2018. Day 2. 10th grade. Problem 7.

Hydroxyapatite is the mineral containing the largest amount of calcium and phosphorous in the human body; it is found in bones and the teeth enamel. It is formed by hydroxide, phosphate and calcium ions. Its composition in mass per cent is 39,9% of calcium and 18,5% of phosphorous.

4.2. Calculate the empirical formula of hydroxyapatite.

Borax is used for the manufacture of enamel, glass and ceramic; it can also be easily transformed in boric acid, which has numerous uses. Borax contains the chemical elements boron, sodium, hydrogen and oxygen; it can be considered a hydrated mixed oxide of sodium and boron. 1,91 g of borax react with 0,0100 mol of hydrochloric acid, forming 1,25 g of boric acid.

4.3. Calculate the empirical formula of borax. $M(\text{H}_3\text{BO}_3) = 61,84 \text{ g}\cdot\text{mol}^{-1}$.

Problem 5. Intermetallic compounds⁵

An intermetallic compound is a compound constituted by two metals. $\text{Cu}(\text{OH})_2$ reacts with $\text{Au}(\text{OH})_3$ in molar ratio 3:1 to form **A**. The infrared spectrum of **A** shows a broad and intense band around the frequency of 3400 cm^{-1} , which indicates the presence of O-H bonds. 0,541 g (1 mmol) of **A** is decomposed in a hydrogen current at 500°C to produce **B** and 0,162 g of water. The elemental analysis of the intermetallic compounds **B** revealed a 50,8% of Au.

5.1. How many moles of water are produced?

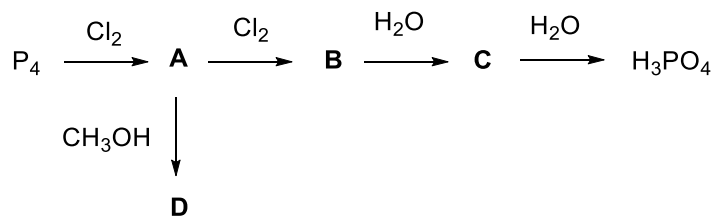
5.2. Using the previous information find the formula of **A**.

5.3. Find the formula of **B**.

⁵ England Chemistry Olympiad, 2002. Problem 4.

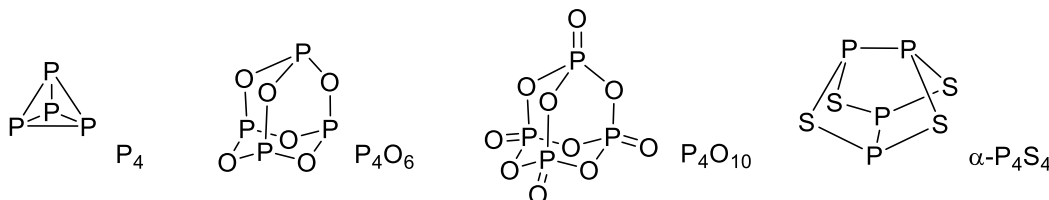
Problem 6. Phosphor and its compounds⁶

The next diagram shows some reactions of phosphorous and its compounds.



- 6.1.** Which substances are represented by letters **A**, **B**, **C** and **D**.
- 6.2.** Though the most stable allotropic modification of this element is P_4 (white phosphorous or red phosphorous), in gas phase is possible to detect a diphosphor molecule. Predict its magnetic properties according to the Theory of Molecular Orbitals.
- 6.3.** Phosphor(V) oxide, P_4O_{10} , is a strong dehydrating reagent that reacts with nitric acid to form metaphosphoric acid, HPO_3 and a white ionic solid. Write the chemical equation corresponding to this reaction.
- 6.4.** Phosphine, PH_3 , is a highly toxic compound, it can be obtained through the disproportion of tetraphosphorous in the presence of potassium hydroxide. Write the equation of the reaction if it is known that in potassium dihydrogenhypophosphite, KH_2PO_2 , is formed.

Phosphorous is a chemical element known by its capability to form molecules and inorganic ions of great structural complexity when combined with non-metallic elements. In such substances it is common to find cycles of different ring sizes in which the phosphorous atoms are alternated with atoms of another chemical elements.



- 6.5.** Draw the Lewis' formula for the species diphosphate, $\text{P}_2\text{O}_7^{4-}$; hypophosphate, $\text{P}_2\text{O}_6^{4-}$; and hypophosphorous acid, H_3PO_2 .
- 6.6.** Suggest Lewis' structures for species $\text{P}_3\text{O}_9^{3-}$ and $(\text{NPCl}_2)_3$ if in both molecules are symmetric and contain a six-atom cycle.

⁶ National Chemistry Contest. Cuba, 2018. Day 2. 10th grade. Problem 6.

Problem 7. Sulfur compounds⁷

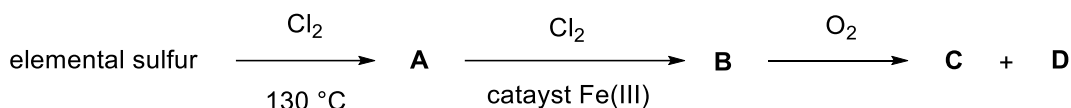
Identify compounds **A** - **D** in the scheme underneath and represent their structures by adequate formulas. Use the additional information offered:

A is a yellow liquid containing 52,5% of Cl and 47,5% of S;

B is a red liquid sensible to humidity;

C is a colorless liquid containing 59,6% of Cl, 26,95% of S and 13,45% of O;

D has a molar mass of $134,96 \text{ g}\cdot\text{mol}^{-1}$ and also can be obtained by direct reaction of **C** with oxygen.



Problem 8. Magnesium compounds⁸

Magnesium is one of the most important elements in the human being. Hundreds of biochemical reactions related to the energetic metabolism and the restauration of DNA are driven by magnesium. Over 300 different enzymes depend on magnesium to exert their catalytic activities. Magnesium keeps blood pressure under control and relax arteries and veins. Magnesium deficient provokes the physiological decay of cells and enable the emergence of cancer. Among many magnesium-containing diet supplements available, the most commonly used is magnesium oxide. Ignited magnesium is difficult to quench because it burns under water, carbon dioxide and nitrogen.

8.1. Write the balance chemical equations for the formation of magnesium oxide by reactions of magnesium with oxygen and carbon dioxide.

8.2. Magnesium hydroxide is formed by the reaction of the metal or the oxide with water. Write the corresponding chemical equations.

When metallic magnesium is heated under N_2 atmosphere, white-yellow compound **A** is formed. Hydrolysis of **A** originates gas **B** which shows basic properties when dissolved in water. The reaction of **B** with hypochlorite ion in aqueous solution generates chloride ion, water and molecular compound **C** which is soluble in water. The reaction of **B** with hydrogen peroxide also produces

⁷ 41th International Chemistry Olympiad. England, 2009. Preparatory Problems. Problem 20.

⁸ 33th International Chemistry Olympiad. Türkiye, 2011. Preparatory Problems. Problem 4.

compound **C** and water. When colorless gas **B** is heated with sodium metal, solid compound **D** and gaseous hydrogen are formed. The reaction of compound **D** with nitrous oxide produces gaseous ammonia, solid sodium hydroxide and a solid compound **E**. When solid **E** is heated, it decomposes in metallic sodium and gaseous nitrogen.

8.3. Write the balanced chemical equations for the formation of compounds **A**, **B**, **C**, **D** and **E**.

8.4. Write the Lewis' structure of the ion present in compound **E**. Select the most stable resonance structure.

Compound **C** was used for the first time as fuel for rockets during the Second Worlds War. Today it is used as low-potency propellant of space crafts. In the presence of certain catalyst as carbon nanofibers or molybdenum nitride supported on alumina, one of the decomposition reactions of **C** involves the production of ammonia and gaseous nitrogen.

8.5. Write the balanced chemical equation for the decomposition reaction of **C** in ammonia and gaseous nitrogen.

8.6. Estimate the energy associated to the decomposition of **C** and the standard enthalpy of formation of NH_3 at 298 K. The standard enthalpy of formation of gaseous and liquid **C** at 298 K are 50,6 y 95,4 $\text{kJ}\cdot\text{mol}^{-1}$, respectively. The mean bond energies at 298 K for $\text{N}\equiv\text{N}$, $\text{N}=\text{N}$, $\text{N}-\text{N}$ and $\text{N}-\text{H}$ are 946, 418, 163 and 389 $\text{kJ}\cdot\text{mol}^{-1}$, respectively.

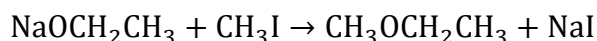
In an experiment, 2,00 mL of **C** were placed in a reaction container previously vacuumed of 1,00 L of capacity that contained an adequate catalyst at 298 K. After the decomposition, the container was cooled to 298 K.

8.7. Calculate the final pressure inside the container. The density of liquid **C** is $1,0045\text{ g}\cdot\text{cm}^{-3}$.

8.8. Calculate the supposing that an isothermal expansion occurs in the container against the atmospheric pressure (1 atm).

Problem 9. Methoxyethane formation⁹

Kinetic data from a study of the reaction between sodium ethoxide and methyl iodide in ethanol at 298 K. It is known that this reaction has an Arrhenius-type behavior.



Experiment	$c_0(\text{NaOEt})/\text{mol}\cdot\text{L}^{-1}$	$c_0(\text{MeI})/\text{mol}\cdot\text{L}^{-1}$	$v_0/\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
1	1,0	2,0	0,00245
2	2,5	0,5	0,00153
3	2,0	1,5	0,00366
4	3,0	2,0	0,00734

9.1. Calculate the partial reaction orders with respect to sodium ethoxide and methyl iodide.

9.2. Calculate the rate constant at 298 K.

If you cannot solve the previous item, consider the rate constant $k = 5,0\cdot 10^{-3} \text{ u}$ to solve the rest of the problem.

9.3. Calculate the activation energy E_A if the reaction rate is increased 4,8 times when the temperature rises up to 313 K.

If you cannot solve the previous item, consider that the activation energy is $E_A = 1,0\cdot 10^5 \text{ J}\cdot\text{mol}^{-1}$ to solve the rest of the problem.

9.4. Calculate the pre-exponential factor A .

Problem 10. Obtaining salts¹⁰

Potassium nitrate and sodium chloride may be obtained through an exchange reaction from potassium chloride and sodium nitrate due to the different solubility of the four salts in water.

10.1. Formulate the chemical equation corresponding to the reaction that happens.

10.2. Plot in the same graph the solubility curves corresponding to the previous salts. Use the data about solubility coefficient in the table.

⁹ National Chemistry Olympiad. Cuba, 2017. Day 1. Common exam. Problem 2.

¹⁰ Vega M., R. V.; Bobes R., M. del P; Fernández N., O., González M., H.; Laboratory No.6 Synthesis of potassium nitrate and sodium chloride. In Laboratory of General Chemistry; Félix Varela, La Habana, Cuba 2007, 66-69. (Adaptation)

	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
KCl	28	34	40	46	51	57
KNO ₃	13	32	64	110	138	202
NaCl	36	36	37	37	38	40
NaNO ₃	73	88	104	124	148	180

10.3. What thermodynamic magnitude determines the solubility dependence on the temperature?

The procedure for a laboratory session carried out in the Faculty of Chemistry of the University of Havana is described below.

1- Dissolve 28 g of sodium nitrate and 24,5 g of potassium chloride with the minimum possible amount of water.

2- Heat to boiling for a minute.

3- Filter in hot and collect the solid sodium chloride that is obtained.

4- Cool the liquid filtrate down to 10 °C.

5- Filter under reduced pressure. Store the potassium nitrate.

6- Add to the liquid in the previous step the sodium chloride previously obtained and heat again until boiling. Boil the solution for 3 min.

7- Filter in hot. Transfer the crystals of sodium chloride to a porcelain capsule and dry them at Bunsen flame. Allow to cool to room temperature and weight the product.

8- Cool the filtrate down to 10 °C.

9- Filter under reduced pressure. Collect the liquid filtrate, measure its temperature and weigh it.

10- Gather both portions of potassium nitrate and recrystallize. Separate the liquid filtrate, measure the temperature and weigh it.

11- Carry out the test to detect chloride ions. If positive, wash the solid with several small portions of iced water until total elimination of chloride ions.

12- Dry the potassium nitrate crystals with towel paper and weigh them.

10.4. Calculate the maximum masses of the products that may be obtained.

10.5. Why the potassium nitrate crystals cannot be dried by heating?

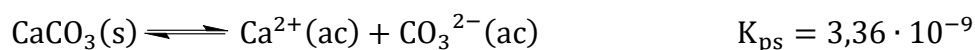
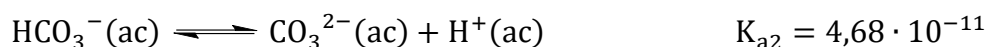
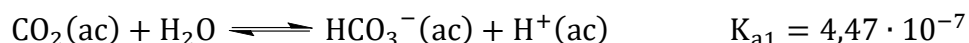
10.6. Why are the masses and the temperatures of the solution determined?

10.7. Calculate the approximate volume of water needed to dissolve all the sodium nitrate and the potassium chloride at a 100 °C.

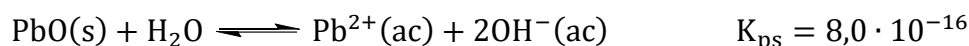
10.8. How can you detect the presence of chloride ions in the crystals of potassium nitrate?

Problem 11. Solubility calculations

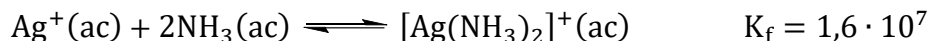
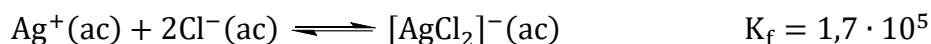
11.1. Calculate the solubility of CaCO_3 in: a) pure water, b) aqueous solution at pH = 3.



11.2. Calculate the solubility of PbO in an aqueous solution: a) at pH = 13,0, b) at pH = 5,0.



11.3. Calculate the solubility of AgCl in: a) aqueous solution of KCl $0,100 \text{ mol}\cdot\text{L}^{-1}$; b) aqueous solution NH_3 $0,100 \text{ mol}\cdot\text{L}^{-1}$.



11.4. Is it possible the quantitative separation of metals nickel, copper and cobalt in the form of sulfides? Suppose an acid solution containing ions Ni^{2+} , Cu^{2+} and Co^{2+} , all to a concentration of $0,010 \text{ mol}\cdot\text{L}^{-1}$. Consider that a separation is quantitative when 99,99% of the target analyte is separated. $K_{ps}(\text{NiS}) = 4 \cdot 10^{-20}$; $K_{ps}(\text{CuS}) = 8 \cdot 10^{-37}$; $K_{ps}(\text{CoS}) = 5 \cdot 10^{-22}$.

11.5. Is it possible to convert lead(II) sulfate into chromate by treatment with a solution of potassium chromate? If possible, calculate the minimum concentration of potassium chromate required for triggering the transformation. $K_{ps}(\text{PbSO}_4) = 1,6 \cdot 10^{-8}$; $K_{ps}(\text{PbCrO}_4) = 2,0 \cdot 10^{-14}$.

Problem 12. Precipitation equilibrium¹¹

The solubility product constant describes the equilibrium that is established when a sparingly soluble electrolyte is dissolved in water. There are different methods to determine the value of this constant. Three examples are illustrated underneath.

A- Calcium oxide was added to pure water and the resultant suspension was allowed to stand for a week in a closed, plastic container, under inert atmosphere. The suspension was filtered to eliminate the solid and stored for later analysis. Immediately, several aliquots of 20,00 mL of the liquid filtrate were titrated with previously standardized hydrochloric acid (concentration $0,0150 \text{ mol}\cdot\text{L}^{-1}$) in the presence of phenolphthalein as acid-base indicator. The volumes of titrant consumed were 30,5; 29,4; 29,8 and 29,6 mL.

12.1. In the previously described procedure several precautions were taken in order to avoid the introduction of errors. Why is it necessary to store the problem solution in a plastic container, under inert temperature? Why the solution was allowed to stand for a week?

12.2. Describe the working principle of acid-base indicators.

12.3. Before carrying out the titration, it is necessary to standardize the hydrochloric acid because this substance is not a primary standard. Which characteristics possess primary standards?

12.4. Calculate the solubility product constant for calcium hydroxide.

B- A 100,0 mL aliquot from a saturated lead(II) solution was reacted with 50 mL of a $2 \text{ mol}\cdot\text{L}^{-1}$ potassium sulfate solution. The mixture was allowed to stand under inert atmosphere. Later, the mixture was filtrated, and the precipitate was washed with cold water, dried under heating and weighted until constant mass. This procedure was repeated, and the results were 0,4821; 0,4822; 0,4827 and 0,4825 g of precipitate.

12.5. Why it is necessary heating the resultant precipitate?

12.6. What is the name of the method employed in this case?

12.7. Calculate the solubility product constant of lead(II) chloride.

$M(\text{PbSO}_4) = 303,26 \text{ mol}\cdot\text{L}^{-1}$, $K_{\text{ps}}(\text{PbSO}_4) = 1,6\cdot 10^{-8}$.

¹¹ National Chemistry Contest. Cuba, 2017. Day 2. 12th grade. Problem 13.

C- Different volumes of $0,100 \text{ mol}\cdot\text{L}^{-1}$ silver nitrate solution and $0,100 \text{ mol}\cdot\text{L}^{-1}$ potassium nitrate were mixed in a beaker. The resultant solutions were stirred for 15 minutes and the redox potentials were measured. For the determination a digital multimeter was employed, a silver electrode (as cathode) as working electrode and a saturated calomel electrode ($E^\circ(\text{Hg}_2\text{Cl}_2/\text{Hg}) = 0,241 \text{ V}$) as reference electrode. To carry out the measurements the working electrode was dropped into the different solutions and the reference electrode was placed in a Luigi's capillary filled with a saturated solution of potassium nitrate (salt bridge) in contact to the problem solution. The redox potential of the electrochemical cell at 25°C are summarized in the table.

Mixture	V(AgNO_3)/mL	V(KCl)/mL	$\Delta E/\text{V}$
1	100	0	0,499
2	90	10	0,493
3	80	20	0,486
4	70	30	0,475
5	60	40	0,457
6	50	50	0,269
7	40	60	0,082
8	30	70	0,064
9	20	80	0,057
10	10	90	0,046

12.8. Why the reference electrode cannot be dropped directly into the problem solution?

12.9. Find mathematic linear equations for the dependency of the cell redox potential when AgNO_3 is in excess (ΔE vs $\ln c(\text{Ag}^+)$) and when KCl is in excess (ΔE vs $\ln c(\text{Cl}^-)$).

12.10. Calculate the standard electrode potential $E^\circ(\text{AgCl}/\text{Ag})$ y $E^\circ(\text{Ag}^+/\text{Ag})$.

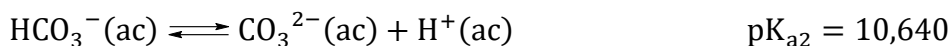
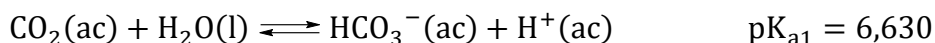
12.11. Calculate the solubility product constant of silver chloride.

Problem 13. Hard water¹²

In Denmark the soil contains mainly limestone. Calcium carbonate from limestone dissolves forming calcium hydrogencarbonate in contact with subterraneous water containing carbon dioxide. As result, subterraneous water is "hard" and when used as topwater the high content of calcium hydrogencarbonate causes problems because of precipitation, for example, in the bathroom and in the kitchen.

¹² 32th International Chemistry Olympiad. Denmark, 2000. Theoretical exam. Problem 6.

Carbon dioxide, CO_2 , is a diprotic acid in aqueous solution. pK_a values at 0°C are:



The volume change in the aqueous phase associated to the dissolution of CO_2 can be neglected when solving the following questions. Consider that the temperature is 0°C .

13.1. The total concentration of carbon dioxide in water is $0,0752 \text{ mol}\cdot\text{dm}^{-3}$ when it is saturated at a partial pressure of carbon dioxide of 1,00 bar. Calculate the volume of gaseous carbon dioxide that can be dissolved in one liter of water under these conditions.

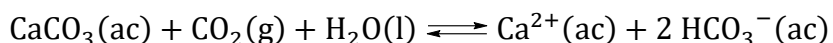
13.2. Calculate the equilibrium concentration of hydrogen ions and the equilibrium concentration of CO_2 in water saturated with carbon dioxide with a partial pressure of 1,00 bar.

13.3. Calculate the equilibrium concentration of hydrogen ions in an aqueous solution of 0,0100 M sodium hydrogencarbonate saturated with carbon dioxide at 1,00 bar partial pressure of carbon dioxide.

13.4. Calculate the equilibrium concentration of hydrogen ions in an aqueous solution of 0,0100 M sodium carbonate saturates with carbon dioxide at 1,00 bar partial pressure of carbon dioxide. Ignore the effects of water dissociation.

13.5. Solubility of calcium carbonate in water at 0°C is 0,0012 g per 100 cm^3 of water. Calculate the concentration of calcium ions in a saturated solution of calcium carbonate in water.

Subterraneous water in Denmark is formed when water contacts the limestone un the soil, and the reaction between limestone and the carbon dioxide dissolve in water takes part:



The equilibrium constant, K , for this reaction is $10^{-4,25}$ at 0°C .

13.6. Calculate the concentration of calcium ions in water in equilibrium with calcium carbonate in an atmosphere with 1,00 bar partial pressure of carbon dioxide.

13.7. A 0,015 M solution of calcium hydroxide is saturated with gaseous carbon dioxide with a partial pressure of 1,00 bar. Calculate the concentration of calcium ions in the solution considering the previous equilibrium equation and the result from 13.6.

13.8. The solution of calcium hydroxide in item 13.7 is diluted with water up to double volume before saturating with carbon dioxide at a partial pressure of 1,00 bar. Calculate the concentration of calcium ions in the resultant solution saturated with CO₂.

13.9. Calculate the solubility product constant for calcium carbonate with the previous data.

Problem 14. Iodine¹³

Iodine is an element essential for life which is found in the living organism in small amounts. When heated to the atmospheric pressure (101,3 kPa) it sublimates, that is, it changes aggregation state from solid to to vapor.



14.1. Calculate the sublimation temperature of iodine.

Species	$\Delta H_f^\circ (\text{kJ} \cdot \text{mol}^{-1})$	$S^\circ (\text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1})$
I ₂ (s)	0,0	116,7
I ₂ (g)	62,3	260,7

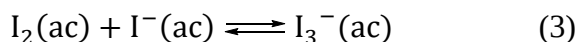
When dichloromethane (CH₂Cl₂) is added to a solution of iodine in water, iodine is distributed between the aqueous phase and the organic phase according to the following equilibrium.



14.2. Calculate the equilibrium constant for the distribution of iodine between water and dichloromethane at 298 K. $\Delta G_f^\circ (\text{kJ} \cdot \text{mol}^{-1})$: I₂(ac) = 16,4; I₂(CH₂Cl₂) = 4,00.

14.3. Why is iodine more soluble in organic substances than in water?

In aqueous solution there is an equilibrium involving the species iodine, iodide and triiodide, according to the following equation.



A method to determine the concentration of I₂ and I₃⁻ in solution is titration with a standard solution of thiosulfate (S₂O₃²⁻). When these iodine species interact with thiosulfate an oxidation-reduction reaction happens giving iodide and tetrathionate (S₄O₆²⁻).

¹³ National Chemistry Contest. Cuba, 2018. Day 2. 11th grade. Problem 8. (reserve)

14.4. Write the equations for the chemical reactions that happen during the titration of I_2 and I_3^- with thiosulfate. Which is the oxidation number of sulfur in thiosulfate ion?

Aiming to determine the equilibrium constant (3) for triiodide formation, the following experiments were carried out at 298 K.

- 50,0 mL of a KI $0,010 \text{ mol}\cdot\text{L}^{-1}$ aqueous solution were added to 25 mL of a solution of I_2 in CH_2Cl_2 , forming two separate phases, aqueous and organic.
- Using a spectrophotometric method, the concentration of iodine in the organic phase was determined equal to $1,20\cdot 10^{-2} \text{ mol}\cdot\text{L}^{-1}$.
- An aliquot of 25,0 mL from the aqueous phase was taken and mixed with an excess amount of a solution of KI to avoid the evaporation of iodine. Then, the resultant solution was titrated with aqueous $\text{Na}_2\text{S}_2\text{O}_3$ $0,0100 \text{ mol}\cdot\text{L}^{-1}$. The end point of the titration was reached after adding 3,10 mL of titrant.

14.5. Calculate the equilibrium concentrations of I^- , I_2 and I_3^- in the aqueous phase and the equilibrium constant for the formation of triiodide ion (3) at 298 K.

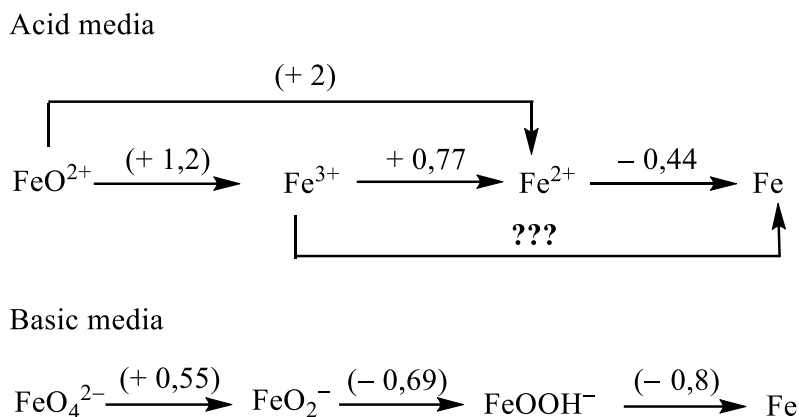
It is possible to calculate the equilibrium constant (3) from electrochemical data. Given the following standard electrode potentials.

$$E^\circ(\text{V}): I_3^-(\text{ac})/I^-(\text{ac}) = 0,536; I_2(\text{ac})/I^-(\text{ac}) = 0,621.$$

14.6. Calculate the equilibrium constant for the formation of triiodide ion (3) at 298 K employing only the potential electrode data.

Problem 15. Iron in solution¹⁴

The Latimer diagram below corresponds to the iron species in aqueous solution in acid and basic media.



15.1. In which conditions is Fe(III) more oxidant? In which conditions is metallic Fe more reductant?

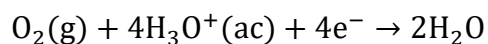
15.2. Does any of iron species disproportionate? Justify.

15.3. May species Fe(III) and Fe coexist in acid media? Justify numerically.

15.4. Calculate the standard redox potential $E^\circ(\text{Fe}^{3+}/\text{Fe})$.

It is possible to comprehend the redox processes happening in natural waters with the aid of electrochemical data. For example, in a lake, cold in the bottom and warm in the surface, the temperature gradient avoids vertical mixing. On the surface the water is partially oxygenated and the iron is present in the form of insoluble particles of $\text{Fe}(\text{OH})_3$. These particles tend to sink. At higher depth, O_2 content is low. If the content of organics or other reductant sources is high enough, the hydroxide is reduced and iron dissolves in the form of Fe^{2+} . Then, the reduce ions diffuse back to the surface where in contact to O_2 are oxidized again to $\text{Fe}(\text{OH})_3$.

15.5. Estimate the approximate potential in the surface of a lake. Consider that the temperature is 25°C , the pressure is 1,01 bar and the pH is 7,0. The composition of air is N_2 78%, O_2 21% and Ar 1% (volume percent).

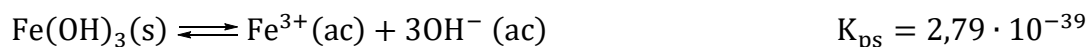


$$E^\circ = 1,23 \text{ V}$$

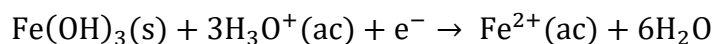
¹⁴ National Chemistry Contest. Cuba, 2018. Day 2. 11th grade. Problem 10. (Reserve)

In fact, the conditions of natural water are slightly different from those calculated by you due to the presence of several dissolved species.

15.6. Calculate the concentrations of Fe^{3+} and Fe^{2+} that exist on the surface of the lake. Consider that the temperature is 25 °C, the pH is 6,5 and the redox potential is 0,65 V.



15.7. Demonstrate that the standard redox potential corresponding to the following process of reduction near the bottom of the lake is $E^{\circ} = 0,97 \text{ V}$.



15.8. Calculate the concentrations of Fe^{3+} and Fe^{2+} that exists in the bottom of the lake. Consider that the temperature is 15 °C, the pH is 6,0 and the potential is 0,15 V. Ignore the effects of the temperature over the equilibrium constants and the standard potentials.

Problem 16. Chlorine¹⁵

Chlorine can be obtained at the laboratory through the reaction between manganese(IV) oxide and hydrochloric acid.

16.1. Write the equation corresponding to the reaction and calculate the equilibrium constant.

Chlorine is obtained at industrial scale by the electrolysis of sodium chloride in aqueous solution in the so-called chloro-alkalis process. Three methods are employed: electrolysis of a cell with mercury amalgam, electrolysis in a cell with diaphragm and electrolysis of a cell with membrane.

In Cuba, chlorine is produced in the factory “Elpidio Sosa” which belongs to the Sagua la Grande Electrochemistry Enterprises, located in the north of Villa Clara province. In this factory the method used is the electrolysis in a cell with mercury cathode. First, the electrolysis of sodium chloride originates in the cathode sodium amalgam $\text{Na}(\text{Hg})$, which is decomposed afterwards when reacted with water.

16.2. Write the chemical equations that happen.

16.3. In your opinion, which is the main inconvenient of this method?

¹⁵ National Chemistry Contest. Cuba, 2017. Day 2. 11th grade. Problem 10.

The electrolysis in a diaphragm cell is mainly used in North America. In this method, anode and cathode are separated to avoid the combination of the gases generated in both of them. The total reaction occurring is



16.4. Write the cathodic and anodic half-reactions.

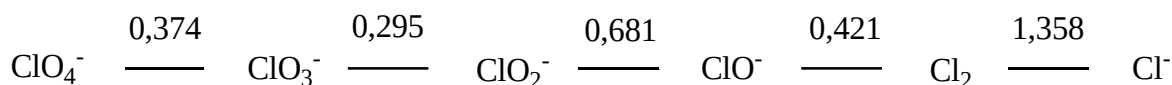
16.5. Calculate the concentration of dissolved sodium chloride. Suppose that a saturated solution of this salt in water is employed (26,0% mass, pH = 7, T = 25 °C, density 1,20 g·mL⁻¹).

16.6. Calculate the minimum potential difference required to force the reaction.

16.7. Write the half-reactions corresponding to other processes that might happen in the cathode and the anode. Reason why they don't happen.

16.8. What is the minimum amount of energy (in kW·h) needed to obtain 1,42 kg of chlorine (M = 70,9 g·mol⁻¹)? Assume an efficiency of 100%.

Given the diagram of potential electrodes of chlorine species in basic media



16.9. Which species disproportionate in basic media?

16.10. Sodium hypochlorite is used for the purification of water. Write the chemical equation of a reaction used to obtain this substance.

E°(acid media)/V: MnO₂/Mn²⁺ = 1,23; O₂/H₂O = 1,23; H⁺/H₂ = 0; Na⁺/Na = -2,71.

Problem 17. Aromaticity¹⁶

Aromaticity is a very important concept in organic chemistry. Compounds having aromatic rings exhibit characteristic physical-chemical properties and reactivity. Hückel's rule allows the identification of aromatic structures. This rule postulates that a cyclic conjugated system is aromatic if the number of p electrons engaging the delocalization of π bonds is equal to $4n + 2$, where n is a non-negative integer. The rule can be extended to polycyclic systems or fused rings.

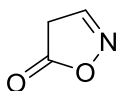
17.1. Provide examples of aromatic structures for $n = 0, 1, 2$.

¹⁶ 40th International Chemistry Olympiad, Hungary, 2008. Preparatory Problems. Problem 15. (Fragments)

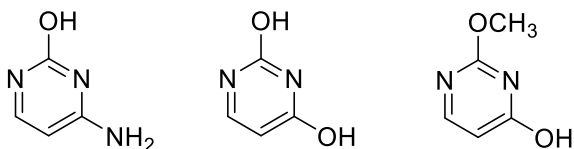
The obtention of aromaticity can be a very favorable driving force in chemical processes, though there are exceptions. Consider the following examples:

17.2. Cyclopentadiene has $pK_a = 18$ while cyclopentene has $pK_a = 45$. Explain the difference.

17.3. Tautomerism can also result in aromatic structures. Show how the following compound can originate an aromatic structure.

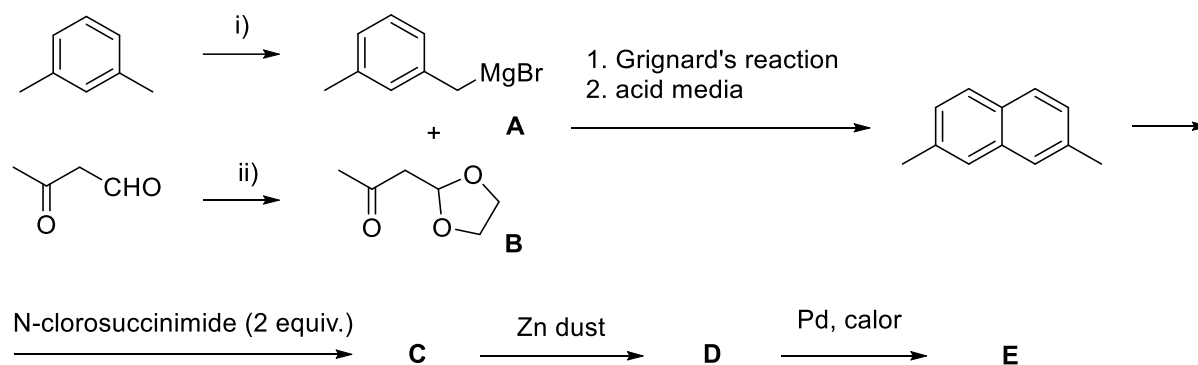


17.4. Nevertheless, some of the flossing aromatic structures prefer to arrange in a tautomer with no aromatic ring. Find the structures. In what context this non-aromatic structures are essential? Provide two additional examples of tautomerism in which the molecules rather missing their aromatic character.



Problem 18. Synthesis of an aromatic hydrocarbon¹⁷

2,7-dimethylnaphthalene can be prepared by reaction between Grignard's reagent **A** and acetal **B**.



18.1. Suggest reaction conditions for the preparation of **A** and **B**.

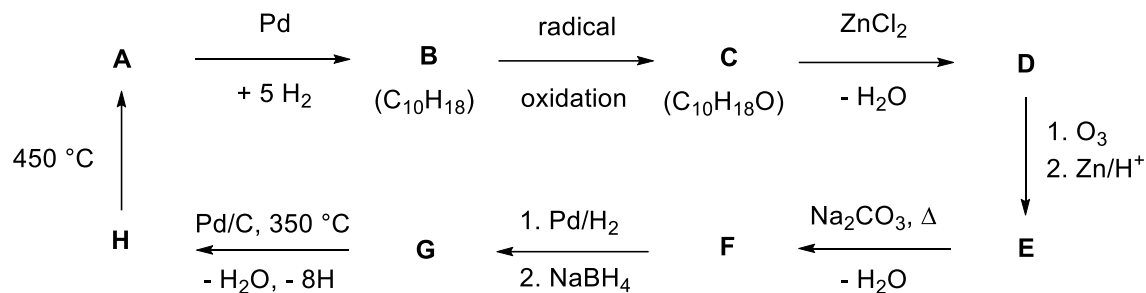
18.2. Explain through a mechanism the formation of 2,7-dimethylnaphthalene.

¹⁷ 40th International Chemistry Olympiad, Hungary, 2008. Preparatory Problems. Problem 19.

18.3. 2,7- dimethylnaphthalene is converted into **E** by the indicated reactions. **E** is a compound with molecular formula $C_{24}H_{12}$. Identify **C**, **D** and **E**.

Problem 19. Two isomers, one synthetic route¹⁸

Find the structure of compounds **A** – **H**, based on the information given in the following scheme.



Data:

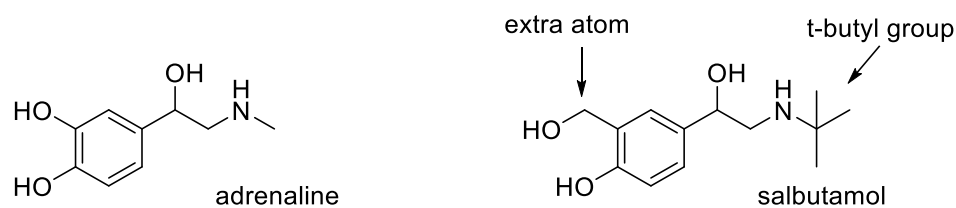
- **A** is a well-known aromatic hydrocarbon.
- A solution of **C** in hexane reacts with sodium (it is possible to see gas evolution), but **C** does not react with chromic acid.
- ^{13}C -RMN analysis shows that **D** and **E** contain only two types of CH_2 groups.
- When a solution of **E** is heated with sodium carbonate, initially an unstable compound is formed, which dehydrates to form **F**.

Problem 20. Salbutamol¹⁹

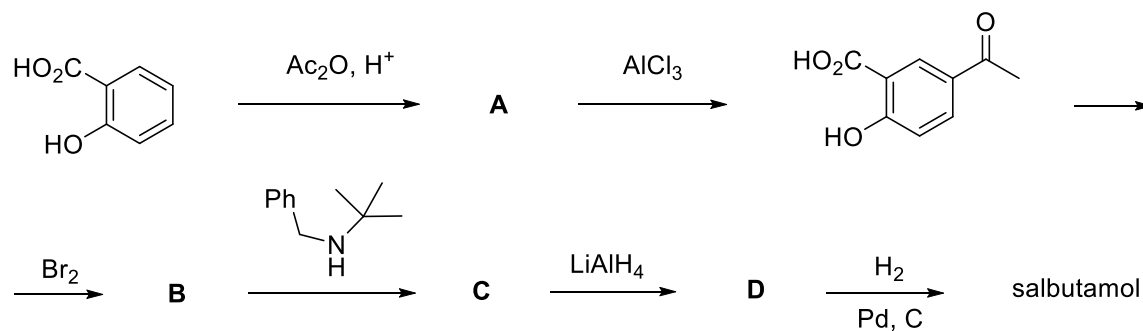
Antiasthma drugs work by dilating the respiratory conducts in lungs, relaxing the constriction that is characteristic of the disease. Salbutamol provokes this effect by imitation the action of adrenaline (epinephrine) hormone. Adrenaline has other effects (cardiac frequency increases) but the chemist from Glaxo company found that the addition of an extra carbon atom avoided side effects harmful for the heart. Besides, the introduction of a tertbutyl group increases the stability of the drug and prologues its effect.

¹⁸ 40th International Chemistry Olympiad, Hungary, 2008. Theoretical exam. Problem 2.

¹⁹ National Chemistry Contest. Cuba 2014. Day 2. 12th grade. (Reserve)



In the scheme below the synthetic sequence used to obtain salbutamol is shown.



20.1. Draw the structures of **A**, **B**, **C** and **D**.

20.2. Which is the common name of a drug that contains compound **A** as active principle? Write the mechanism of the reaction of **A** with AlCl_3 (Fries' transposition).

20.3. Draw the mechanism of the bromination reaction.

20.4. Why is the benzylated derivative used instead of simple tertbutylamine?

ANSWERS

Problem 1

1.1.

$$n(A) = \frac{PV}{RT} = \frac{101,0 \cdot 0,05325}{8,314 \cdot 298,15} = 0,00217 \text{ mol}$$

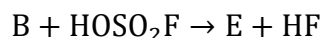
$$m(A) = \frac{100 - 36,7}{100} \cdot 0,450 = 0,285 \text{ g}$$

Gas A can be monoatomic, diatomic ...

$$M(A) = \frac{m(A)}{n(A)a} = \frac{0,285}{0,00217 \cdot a} = \frac{131}{a}$$

For $a = 1$, $M(A) = 131 \text{ g} \cdot \text{mol}^{-1}$ which corresponds to Xe

1.2.



As the reaction is stoichiometric, B also contains two fluor atoms.

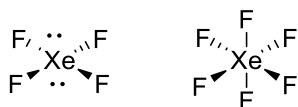
$$\frac{n(\text{F})}{n(\text{Xe})} = \frac{m(\text{F})m(\text{Xe})}{M(\text{F})m(\text{Xe})} = \frac{\omega(\text{F})m(\text{Xe})}{M(\text{F})\omega(\text{Xe})}$$

For C it is true that $n(\text{F}):n(\text{Xe}) = 4$.

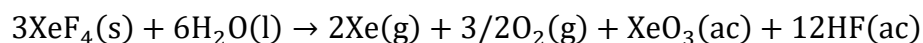
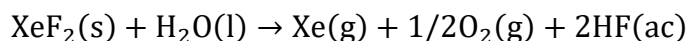
For D it is true that $n(\text{F}):n(\text{Xe}) = 6$.

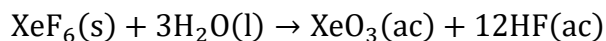


1.3.



1.4.

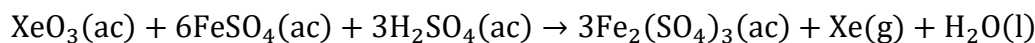




1.5.

$$n(\text{total}) = \frac{PV}{RT} = \frac{100 \cdot 0,0602}{8,314 \cdot 290} = 0,00250 \text{ mol}$$

$$n(\text{O}_2) = 0,4 \cdot 0,00250 = 0,00100 \text{ mol} \quad n(\text{Xe}) = 0,00150 \text{ mol}$$



$$n(\text{XeO}_3) = \frac{1}{6} \cdot n(\text{FeSO}_4) = \frac{1}{6} \cdot n(\text{FeSO}_4) = 0,0036 = 0,0006 \text{ mol}$$

$$n(\text{O}_2) = 0,5n(\text{XeF}_2) + 0,5n(\text{XeF}_4)$$

$$n(\text{Xe}) = n(\text{XeF}_2) + 2/3n(\text{XeF}_4)$$

$$n(\text{XeO}_3) = 1/3n(\text{XeF}_4) + n(\text{XeF}_6)$$

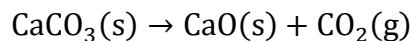
After solving the previous equation system

$$n(\text{XeF}_2) = 0,0005 \text{ mol} \quad n(\text{XeF}_4) = 0,00150 \text{ mol} \quad n(\text{XeO}_3) = 0,0001 \text{ mol}$$

$$\chi(\text{XeF}_2) = 23,8\% \quad \chi(\text{XeF}_4) = 71,4\% \quad \chi(\text{XeO}_3) = 4,8\%$$

Problem 2

2.1.



2.2.

$$\text{A } 900^\circ\text{C}, \quad m(\text{CO}_2) = 3,06 - 2,03 = 1,03 \text{ g}$$

$$n(\text{CaCO}_3) = n(\text{CaC}_2\text{O}_4) = n(\text{CO}_2) = \frac{m(\text{CO}_2)}{M(\text{CO}_2)} = \frac{1,03}{44,0} = 0,0234 \text{ mol}$$

$$m(\text{CaCO}_3) = n(\text{CaCO}_3)M(\text{CaCO}_3) = 0,0234 \cdot 100 = 2,34 \text{ g}$$

$$m(\text{CaC}_2\text{O}_4) = n(\text{CaC}_2\text{O}_4)M(\text{CaC}_2\text{O}_4) = 0,0234 \cdot 128 = 3,00 \text{ g}$$

$$\text{A } 500^\circ\text{C}, \quad m(\text{MgO}) = 3,06 - 2,34 = 0,72 \text{ g}$$

$$n(\text{MgC}_2\text{O}_4) = n(\text{MgO}) = \frac{m(\text{MgO})}{M(\text{MgO})} = \frac{0,72}{40,0} = 0,018 \text{ mol}$$

$$m(\text{MgC}_2\text{O}_4) = n(\text{MgC}_2\text{O}_4)M(\text{MgC}_2\text{O}_4) = 0,018 \cdot 112 = 2,02 \text{ g}$$

$$\omega(\text{MgC}_2\text{O}_4) = \frac{2,02}{2,02 + 3,00} = 40\% \quad \omega(\text{CaC}_2\text{O}_4) = 60\%$$

2.3.

Elimination of water.

2.4.

$$\text{A } 900^\circ\text{C}, \quad n(\text{CO}_2) = n(\text{CaO}) = \frac{m(\text{CaO})}{M(\text{CaO})} = \frac{2,80}{56,0} = 0,0500 \text{ mol}$$

$$M(\text{CO}_2) = \frac{m(\text{CO}_2)}{n(\text{CO}_2)} = \frac{5,00 - 2,80}{0,0500} = 44,0 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{C}) = M(\text{CO}_2) - 2M(\text{O}) = 44,0 - 2 \cdot 16,0 = 12,0 \text{ g} \cdot \text{mol}^{-1}$$

2.5.

$$n(\text{B}_2\text{O}_3) = \frac{m(\text{B}_2\text{O}_3)}{M(\text{B}_2\text{O}_3)} = \frac{3,5}{69,6} = 0,050 \text{ mol} \quad n(\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} = \frac{4,4 - 3,5}{18,0} = 0,050 \text{ mol}$$

$$X = \text{B}_2\text{O}_3 \cdot \text{H}_2\text{O} = \text{H}_2\text{B}_2\text{O}_4 = \text{HBO}_2$$

Problem 3

3.1.

$$\frac{3}{2} = \frac{n(\text{Si})}{n(\text{Mn})} = \frac{m(\text{Si})M(\text{Mn})}{M(\text{Si})m(\text{Mn})} = \frac{\omega(\text{Si})M(\text{Mn})}{M(\text{Si})\omega(\text{Mn})}$$

$$\frac{\omega(\text{Si})}{\omega(\text{Mn})} = \frac{3M(\text{Si})}{2M(\text{Mn})} = \frac{3 \cdot 28,1}{2 \cdot 54,9} = 0,768$$

$$\omega(\text{Si}) + \omega(\text{Mn}) = 3\% \quad \rightarrow \quad \omega(\text{Si}) = 1,3\% \quad \omega(\text{Mn}) = 1,7\%$$

3.2.

If hf = melted iron and fm = ferromanganese.

It is possible to calculate the amount of iron oxide formed from melted iron.

$$n(\text{FeO}) = n(\text{Fe})_{\text{oxidizes}} = 0,5 \cdot \frac{0,94 \cdot m(\text{hf})}{M(\text{Fe})}$$

Iron oxide reacts with the components of melted iron and ferromanganese.

$$n(\text{FeO}) = n(\text{FeO})_{\text{Mn}} + n(\text{FeO})_{\text{Si}} + n(\text{FeO})_{\text{C}}$$

$$n(\text{FeO}) = n(\text{Mn}) + 2n(\text{Si}) + 2n(\text{C}) = \frac{m(\text{Mn})}{M(\text{Mn})} + 2 \frac{m(\text{Si})}{M(\text{Si})} + 2 \frac{m(\text{C})_{\text{FeO}}}{M(\text{C})}$$

Masses of Mn, Si and C that react with iron oxide are

$$m(\text{Mn}) = 0,017 \cdot m(\text{hf}) + 0,90 \cdot m(\text{fm})$$

$$m(\text{Si}) = 0,013 \cdot m(\text{hf}) + 0,01 \cdot m(\text{fm})$$

$$m(\text{C}) = 0,03 \cdot m(\text{hf}) + 0,06 \cdot m(\text{fm}) - m(\text{C})_{\text{exc}}$$

A portion of carbon is in excess and stays in the final steel. But all the iron in the steel comes from melted iron. Then

$$m(\text{C})_{\text{exc}} = 0,002 \cdot m(\text{steel}) = 0,002 \cdot \frac{0,94 \cdot m(\text{hf})}{0,998}$$

Combining the previous equations

$$0,5 \cdot \frac{0,94 \cdot m(\text{hf})}{M(\text{Fe})} = \frac{m(\text{Mn})}{M(\text{Mn})} + 2 \frac{m(\text{Si})}{M(\text{Si})} + 2 \frac{m(\text{C})}{M(\text{C})}$$

$$\frac{0,5 \cdot 0,94 \cdot m(\text{hf})}{55,8} = \frac{0,017 \cdot m(\text{hf}) + 0,90 \cdot m(\text{fm})}{54,9} + 2 \frac{0,013 \cdot m(\text{hf}) + 0,01 \cdot m(\text{fm})}{28,1} + 2 \frac{0,03 \cdot m(\text{hf}) + 0,06 \cdot m(\text{fm}) - m(\text{C})_{\text{exc}}}{12,0}$$

Substituting $m(\text{hf}) = 1,00$ ton one gets

$$n(\text{FeO}) = 842 \text{ mol}$$

$$m(\text{C})_{\text{exc}} = 0,00188 \text{ ton}$$

$$m(\text{fm}) = 0,0922 \text{ ton} = 92,2 \text{ kg}$$

Problem 4

4.1.

$$\frac{m(\text{Be})}{m(\text{Al})} = \frac{1}{2} = \frac{9,01 \cdot n(\text{Be})}{26,98 \cdot n(\text{Al})} \rightarrow \frac{n(\text{Be})}{n(\text{Al})} \approx \frac{3}{2}$$

If beryl formula is $3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2$

$$\frac{n(\text{Si})}{n(\text{O})} = \frac{1}{3} = \frac{x}{2x + 6} \rightarrow 3x = 6 + 2x \rightarrow x = 6$$

Beryl formula: $3\text{BeO} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$

4.2.

In 100 g hydroxyapatite,

$$n(\text{Ca}^{2+}) = n(\text{Ca}) = \frac{m(\text{Ca})}{A_r(\text{Ca})} = \frac{39,9}{40,08} = 0,996 \text{ mol}$$

$$n(\text{PO}_4^{3-}) = n(\text{P}) = \frac{m(\text{P})}{A_r(\text{P})} = \frac{18,5}{30,97} = 0,597 \text{ mol}$$

From the charge balance one gets

$$2n(\text{Ca}^{2+}) = 3n(\text{PO}_4^{3-}) + n(\text{OH}^-)$$

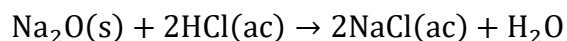
$$n(\text{OH}^-) = 0,201 \text{ mol}$$

$$n(\text{Ca}^{2+}) : n(\text{PO}_4^{3-}) : n(\text{OH}^-) = 0,996 : 0,597 : 0,20 \approx 5 : 3 : 1 = 10 : 6 : 2$$

Hydroxyapatite formula: $10\text{CaO} \cdot 3\text{P}_2\text{O}_5 \cdot \text{H}_2\text{O}$

4.3.

If borax formula is $x\text{Na}_2\text{O} \cdot y\text{B}_2\text{O}_3 \cdot z\text{H}_2\text{O}$



$$n(\text{Na}_2\text{O}) = n(\text{HCl})/2 = 0,0100/2 = 0,00500 \text{ mol}$$

$$M(\text{Na}_2\text{O}) = 2 \cdot A_r(\text{Na}) + A_r(\text{O}) = 2 \cdot 22,99 + 16,00 = 61,98 \text{ g} \cdot \text{mol}^{-1}$$

$$m(\text{Na}_2\text{O}) = n(\text{Na}_2\text{O}) \cdot M(\text{Na}_2\text{O}) = 0,00500 \cdot 61,98 = 0,309 \text{ g}$$

$$n(\text{B}_2\text{O}_3) = \frac{n(\text{H}_3\text{BO}_3)}{2} = \frac{m(\text{H}_3\text{BO}_3)}{2M(\text{H}_3\text{BO}_3)} = \frac{1,24}{2 \cdot 61,84} = 0,0101 \text{ mol}$$

$$M(\text{B}_2\text{O}_3) = 2 \cdot A_r(\text{B}) + 3 \cdot A_r(\text{O}) = 2 \cdot 10,81 + 3 \cdot 16,00 = 69,62 \text{ g} \cdot \text{mol}^{-1}$$

$$m(\text{B}_2\text{O}_3) = n(\text{B}_2\text{O}_3) \cdot M(\text{B}_2\text{O}_3) = 0,0101 \cdot 69,62 = 0,703 \text{ g}$$

$$m(\text{H}_2\text{O}) = m(\text{bórax}) - m(\text{Na}_2\text{O}) - m(\text{B}_2\text{O}_3) = 1,91 - 0,309 - 0,703 \approx 0,90 \text{ g}$$

$$M(\text{H}_2\text{O}) = 2 \cdot A_r(\text{H}) + A_r(\text{O}) = 2 \cdot 1,01 + 16,00 = 18,02 \text{ g} \cdot \text{mol}^{-1}$$

$$n(\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} = \frac{0,90}{18,02} = 0,0499 \text{ mol}$$

$$n(\text{Na}_2\text{O}):n(\text{B}_2\text{O}_3):n(\text{H}_2\text{O}) = 0,00500:0,0101:0,0499 \approx 1:2:10$$

$$\text{Formula of borax: } \text{Na}_2\text{O} \cdot 2\text{B}_2\text{O}_3 \cdot 10\text{H}_2\text{O}$$

Problem 5

5.1.

$$n(\text{H}_2\text{O}) = \frac{m(\text{H}_2\text{O})}{M(\text{H}_2\text{O})} = \frac{0,162}{18,0} = 0,009 \text{ mol}$$

5.2.

Each mol of A contains 9 mol of oxygen, probably in the form of OH.

The stoichiometric ration must be 3: 1 in the product.

$$M(\text{A}) = 541 \text{ g} \cdot \text{mol}^{-1}$$

$$9M(\text{OH}) + 3M(\text{Cu}) + M(\text{Au}) = 9 \cdot 17 + 3 \cdot 63,5 + 197 = 541 \text{ g} \cdot \text{mol}^{-1}$$

$$\text{A} = \text{AuCu}_3(\text{OH})_9$$

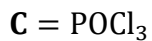
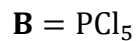
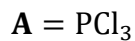
5.3.

$$\frac{n(\text{Cu})}{n(\text{Au})} = \frac{m(\text{Cu})M(\text{Au})}{M(\text{Cu})m(\text{Au})} = \frac{\omega(\text{Cu})M(\text{Au})}{M(\text{Cu})\omega(\text{Au})} = \frac{49,2 \cdot 197}{63,5 \cdot 50,8} = 3$$

$$\text{B} = \text{AuCu}_3$$

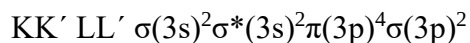
Problem 6

6.1.

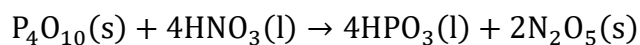


6.2.

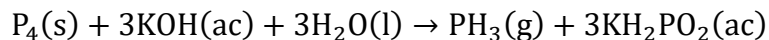
The electronic structure of P_2 molecule is



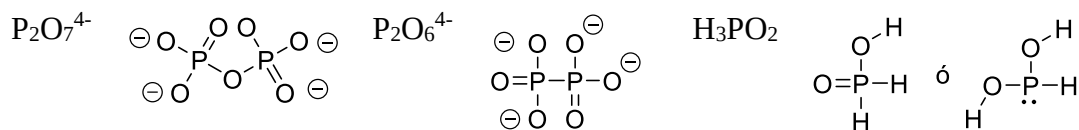
6.3.



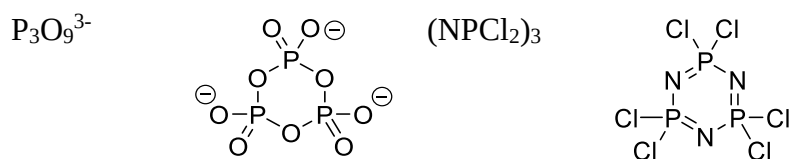
6.4.



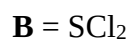
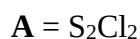
6.5.



6.6.



Problem 7



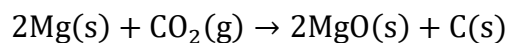
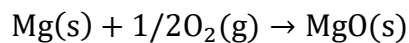
$$\omega(\text{Cl}, \text{A}) = \frac{2M(\text{Cl})}{2M(\text{Cl}) + 2M(\text{S})} = \frac{2 \cdot 35,5}{2 \cdot 35,5 + 2 \cdot 32} = 52,6\%$$

$$\omega(\text{Cl}, \text{C}) = \frac{2M(\text{Cl})}{2M(\text{Cl}) + M(\text{S}) + M(\text{O})} = \frac{2 \cdot 35,5}{2 \cdot 35,5 + 32 + 16} = 59,7\%$$

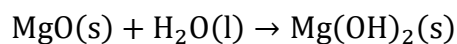
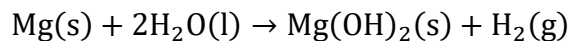
$$M(\text{D}) = M(\text{S}) + 2M(\text{O}) + 2M(\text{Cl}) = 32 + 2 \cdot 16 + 2 \cdot 35,5 = 135 \text{ g} \cdot \text{mol}^{-1}$$

Problem 8

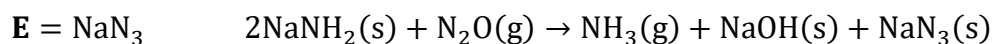
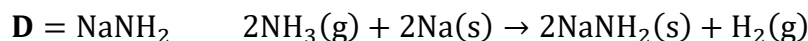
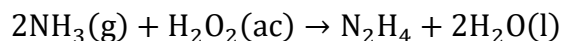
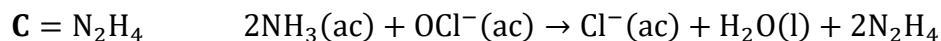
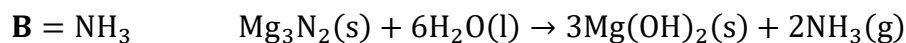
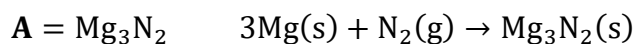
8.1.



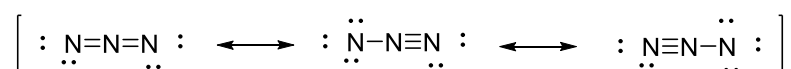
8.2.



8.3.

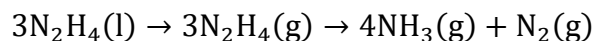


8.4.



The first structure is the most stable.

8.5.



8.6.

$$\Delta H_{\text{vap}}^\circ = \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4, \text{g}) - \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4, \text{l}) = 95,4 - 50,6 = 44,8 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta H_{\text{r}}^\circ = 3\varepsilon(\text{N}-\text{N}) + 12\varepsilon(\text{N}-\text{H}) + 3\Delta H_{\text{vap}}^\circ - \varepsilon(\text{N}\equiv\text{N}) - 12\varepsilon(\text{N}-\text{H})$$

$$\Delta H_{\text{r}}^\circ = 3 \cdot 163 + 3 \cdot 44,8 - 946 = -322,6 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta H_{\text{r}}^\circ = 4 \cdot \Delta H_{\text{f}}^\circ(\text{NH}_3, \text{g}) - 3 \cdot \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4, \text{l})$$

$$-322,6 = 4 \cdot \Delta H_f^\circ(\text{NH}_3, \text{g}) - 3 \cdot 50,6 \quad \rightarrow \quad \Delta H_f^\circ(\text{NH}_3, \text{g}) = -42,7 \text{ kJ} \cdot \text{mol}^{-1}$$

8.7.

$$m = d \cdot V = 2,00 \cdot 1,0045 = 2,00 \text{ g} \quad n(\text{N}_2\text{H}_4) = \frac{m(\text{N}_2\text{H}_4)}{M(\text{N}_2\text{H}_4)} = \frac{2,00}{32,0} = 0,0625 \text{ mol}$$

$$n(\text{total}) = 2 \cdot 0,0625 = 0,125 \text{ mol}$$

$$P(\text{total}) = \frac{n(\text{total})RT}{V} = \frac{0,125 \cdot 8,314 \cdot 298}{1,00} = 310 \text{ kPa}$$

8.8.

$$V(\text{final}) = \frac{n(\text{total})RT}{P} = \frac{0,104 \cdot 8,314 \cdot 298}{101,3} = 2,55 \text{ L}$$

$$w = P \cdot \Delta V = 101,3 \cdot (2,55 - 1,00) = 157 \text{ J}$$

Problem 9

9.1.

$$\text{Sea } v_r = k \cdot c^x(\text{NaOEt}) \cdot c^y(\text{MeI})$$

$$\text{Exp. 4} \quad 0,00734 = k \cdot (3,0)^x \cdot (2,0)^y$$

$$\text{Exp. 1} \quad 0,00245 = k \cdot (1,0)^x \cdot (2,0)^y$$

$$3,00 = (3,0)^x \quad x = 1$$

$$\text{Exp. 3} \quad 0,00366 = k \cdot (2,0)^1 \cdot (1,5)^y$$

$$\text{Exp. 1} \quad 0,00245 = k \cdot (1,0)^1 \cdot (2,0)^y$$

$$1,49 = 2 \cdot (0,75)^y \quad 0,745 = (0,75)^y \quad y = 1$$

9.2.

$$k = \frac{v_r}{c^x(\text{NaOEt}) \cdot c^y(\text{MeI})}$$

$$\text{Exp. 1} \quad k = \frac{0,00245}{1,0 \cdot 2,0} = 0,00123 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

$$\text{Exp. 2} \quad k = \frac{0,00153}{2,5 \cdot 0,5} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

$$\text{Exp. 3} \quad k = \frac{0,00366}{2,0 \cdot 1,5} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

$$\text{Exp. 4} \quad k = \frac{0,00734}{3,0 \cdot 2,0} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

$$\bar{k} = 0,00122 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

9.3.

$$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right) \rightarrow \ln\left(\frac{k_2}{k_1}\right) = -\frac{E_A}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

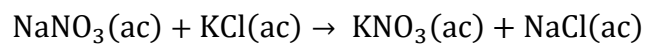
$$E_A = \frac{-R \cdot \ln\left(\frac{k_2}{k_1}\right)}{\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} = \frac{-8,314 \cdot \ln 4,8}{\left(\frac{1}{313} - \frac{1}{298}\right)} = \frac{-13,0}{-1,61 \cdot 10^{-4}} = 8,1 \cdot 10^4 \text{ J} \cdot \text{mol}^{-1}$$

9.4.

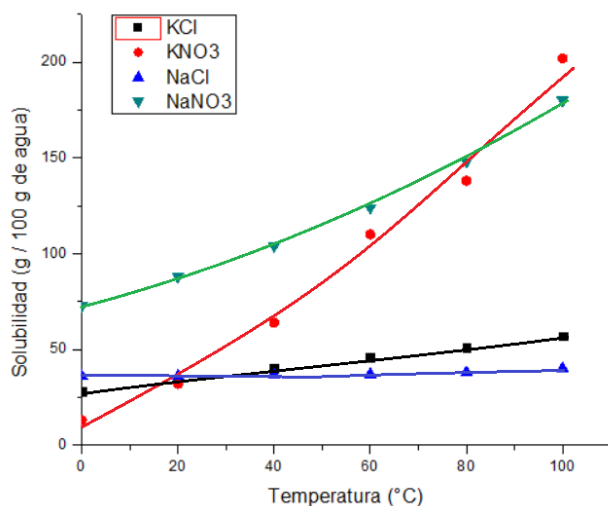
$$A = \frac{k}{\exp\left(-\frac{E_A}{R \cdot T}\right)} = \frac{1,22 \cdot 10^{-3}}{\exp\left(-\frac{8,1 \cdot 10^4}{8,314 \cdot 298}\right)} = \frac{1,22 \cdot 10^{-3}}{6,33 \cdot 10^{-15}} = 1,9 \cdot 10^{11} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

Problem 10

10.1.



10.2.



10.3.

The dissolution enthalpy change.

10.4.

$$n(\text{NaNO}_3) = \frac{m(\text{NaNO}_3)}{M(\text{NaNO}_3)} = \frac{28}{85} = 0,33 \text{ mol} \quad n(\text{KCl}) = \frac{m(\text{KCl})}{M(\text{KCl})} = \frac{24,5}{74,5} = 0,33 \text{ mol}$$

The reactants are in stoichiometric amounts.

$$m(\text{KNO}_3) = n(\text{KNO}_3)M(\text{KNO}_3) = 0,33 \cdot 101 = 33,3 \text{ g}$$

$$m(\text{NaCl}) = n(\text{NaCl})M(\text{NaCl}) = 0,33 \cdot 58,5 = 19,3 \text{ g}$$

10.5.

Potassium nitrate is an explosive.

10.6.

To approximately assess the mass lost due to solubility.

10.7.

$$m(\text{H}_2\text{O}) = \frac{m(\text{NaNO}_3)}{S(\text{NaNO}_3)} \cdot 100 = \frac{28}{88} \cdot 100 = 32 \text{ g} \rightarrow 32 \text{ mL}$$

$$m(\text{H}_2\text{O}) = \frac{m(\text{KCl})}{S(\text{KCl})} \cdot 100 = \frac{24,5}{34} \cdot 100 = 72 \text{ g} \rightarrow 72 \text{ mL}$$

Assuming independent solubilities, 72 mL of water are needed.

10.8.

In a reaction flask a portion of the crystals is dissolved. A few drops of silver nitrated are added. The advent of a white color of AgCl indicates the presence of chloride ions.

Problem 11**11.1.**

In pure water the hydrolysis of carbonate and calcium ions is negligible.

$$c(\text{CaCO}_3) = [\text{Ca}^{2+}] \approx [\text{CO}_3^{2-}] \rightarrow K_{\text{ps}} = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = [\text{Ca}^{2+}]^2$$

$$[\text{Ca}^{2+}] = \sqrt{K_{\text{ps}}} = \sqrt{3,36 \cdot 10^{-9}} = 5,80 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

In acid media hydrolysis of carbonate ion is large.

$$c(\text{CaCO}_3) = [\text{Ca}^{2+}] = [\text{CO}_3^{2-}] + [\text{HCO}_3^-] + [\text{H}_2\text{CO}_3] = [\text{CO}_3^{2-}]/\alpha_2$$

$$\alpha_2 = \frac{K_1 K_2}{[\text{H}^+]^2 + K_1[\text{H}^+] + K_1 K_2} = 2,09 \cdot 10^{-11}$$

$$K_{ps} = [\text{Ca}^{2+}][\text{CO}_3^{2-}] = [\text{Ca}^{2+}]\alpha_2[\text{Ca}^{2+}] = \alpha_2[\text{Ca}^{2+}]^2$$

$$[\text{Ca}^{2+}] = \sqrt{\frac{K_{ps}}{\alpha_2}} = \sqrt{\frac{3,36 \cdot 10^{-9}}{2,09 \cdot 10^{-11}}} = 12,7 \text{ mol} \cdot \text{L}^{-1}$$

At pH = 3 calcium carbonate will dissolve forming carbon dioxide.

11.2.

In basic media the formation of a complex ion is the most important reaction.

$$c(\text{PbO}) = [\text{Pb}^{2+}] + [\text{Pb}(\text{OH})_3]^- \approx [\text{Pb}(\text{OH})_3]^- = \frac{K_a}{[\text{H}^+]} = \frac{10^{-15}}{[10^{-13}]} = 0,010 \text{ mol} \cdot \text{L}^{-1}$$

In acid media the formation of the complex acid is negligible.

$$c(\text{PbO}) = [\text{Pb}^{2+}] + [\text{Pb}(\text{OH})_3]^- \approx [\text{Pb}^{2+}] = \frac{K_{ps}}{[\text{OH}^-]^2} = \frac{8,0 \cdot 10^{-16}}{[1,0 \cdot 10^{-9}]^2} = 800 \text{ mol} \cdot \text{L}^{-1}$$

At pH = 5 lead(II)oxide will dissolve.

11.3.

In the potassium chloride solution

$$c(\text{AgCl}) = [\text{Ag}^+] + [\text{AgCl}_2]^- \approx [\text{AgCl}_2]^- = K_f[\text{Ag}^+][\text{Cl}^-]^2 = K_f K_{ps}[\text{Cl}^-]$$

$$c(\text{AgCl}) = 1,8 \cdot 10^{-10} \cdot 1,7 \cdot 10^5 \cdot 0,100 = 3,1 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

In the ammonia solution

$$c(\text{AgCl}) \approx [\text{Cl}^-] = [\text{Ag}^+] + [\text{Ag}(\text{NH}_3)_2]^+ \approx [\text{Ag}(\text{NH}_3)_2]^+ = K_f[\text{Ag}^+][\text{NH}_3]^2$$

$$K_{ps} = [\text{Ag}^+][\text{Cl}^-] = \frac{[\text{Cl}^-]}{K_f[\text{NH}_3]^2}[\text{Cl}^-] \quad \rightarrow \quad [\text{Cl}^-] = \sqrt{K_{ps}K_f[\text{NH}_3]^2}$$

$$c(\text{AgCl}) \approx [\text{Cl}^-] = \sqrt{K_{ps}K_f[\text{NH}_3]^2} = 0,0054 \text{ mol} \cdot \text{L}^{-1}$$

11.4.

Considering the conditions set in the problem, the only possible quantitative separation to consider is the precipitation of copper ions.

The precipitation of Co^{2+} ions starts when the sulfide concentration is

$$[\text{S}^{2-}] = \frac{K_{\text{ps}}}{[\text{Co}^{2+}]} = \frac{5 \cdot 10^{-22}}{0,010} = 5 \cdot 10^{-20} \text{ mol} \cdot \text{L}^{-1}$$

The concentration of Cu^{2+} ions in this conditions is

$$[\text{Cu}^{2+}] = \frac{K_{\text{ps}}}{[\text{S}^{2-}]} = \frac{8 \cdot 10^{-37}}{5 \cdot 10^{-20}} = 1,6 \cdot 10^{-17} \text{ mol} \cdot \text{L}^{-1}$$

$$\frac{[\text{Cu}^{2+}]}{(\text{Cu}^{2+})_{\text{inicial}}} = \frac{1,6 \cdot 10^{-17}}{0,010} = 1,6 \cdot 10^{-13}\%$$

More than 99,99% of copper ion precipitates, so its quantitative separation is possible.

11.5.

The reaction to consider is



$$K = \frac{K_{\text{ps}}(\text{PbSO}_4)}{K_{\text{ps}}(\text{PbCrO}_4)} = \frac{1,6 \cdot 10^{-8}}{2,0 \cdot 10^{-14}} = 8 \cdot 10^5$$

The high value of the equilibrium constant means that the reaction proceeds in large extension.

In a saturated solution of lead(II)sulfate, the concentration of Cu^{2+} ions is

$$[\text{Pb}^{2+}] = \sqrt{K_{\text{ps}}(\text{PbSO}_4)} = \sqrt{1,6 \cdot 10^{-8}} = 1,3 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

The minimum necessary concentration necessary for the reaction to start is

$$[\text{CrO}_4^{2-}] = \frac{K_{\text{ps}}(\text{CrSO}_4)}{[\text{Pb}^{2+}]} = \frac{2 \cdot 10^{-14}}{1,3 \cdot 10^{-4}} = 1,6 \cdot 10^{-10} \text{ mol} \cdot \text{L}^{-1}$$

Problem 12

12.1.

The container should be plastic for the base don't react with glass. The atmosphere must be inert because the basic solutions absorb carbon dioxide from air. To carry out the analysis there should be enough time to reach equilibrium.

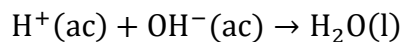
12.2.

An indicator is a weak acid or a weak base in which the molecular species and the ionic species have different colors. Therefore, the resultant color of the solution depends on the media pH.

12.3.

A primary standard is a chemically stable substance, of definite composition, of high molar mass, inert towards the air components and that reacts quantitatively and stoichiometrically.

12.4.



$$\bar{V}(\text{HCl}) = \frac{29,4 + 29,8 + 29,6}{3} = 29,6 \text{ mL}$$

The volume 30,5 mL should not be considered.

$$c(\text{OH}^-) = \frac{n(\text{OH}^-)}{V(\text{Ca}(\text{OH})_2)} = \frac{c(\text{HCl}) \cdot \bar{V}(\text{HCl})}{V(\text{Ca}(\text{OH})_2)} = \frac{0,0150 \cdot 29,6}{20,0} = 0,0222 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{Ca}^{2+}) = \frac{c(\text{OH}^-)}{2} = \frac{0,0222}{2} = 0,0111 \text{ mol} \cdot \text{L}^{-1}$$

$$K_{\text{ps}} = [\text{Ca}^{2+}] \cdot [\text{OH}^-]^2 = 0,0111 \cdot (0,0222)^2 = 5,47 \cdot 10^{-6}$$

12.5.

The precipitate is heated until complete dryness.

12.6.

Gravimetric methods.

12.7.

$$\bar{M}(\text{PbSO}_4) = \frac{0,4821 + 0,4822 + 0,4827 + 0,4825}{4} = 0,4824 \text{ g}$$

$$n(\text{Pb}^{2+}) = n(\text{PbSO}_4) = \frac{m(\text{PbSO}_4)}{M(\text{PbSO}_4)} = \frac{0,4824}{303,26} = 0,001591 \text{ mol}$$

$$c(\text{Pb}^{2+}) = \frac{n(\text{Pb}^{2+})}{V(\text{PbCl}_2)} = \frac{0,001591}{0,1000} = 0,01591 \text{ mol} \cdot \text{L}^{-1}$$

$$c(\text{Cl}^-) = 2 \cdot (\text{Pb}^{2+}) = 2 \cdot 0,01591 = 0,03181 \text{ mol} \cdot \text{L}^{-1}$$

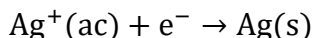
$$K_{\text{ps}} = [\text{Pb}^{2+}] \cdot [\text{Cl}^-]^2 = 0,01591 \cdot (0,03181)^2 = 1,610 \cdot 10^{-5}$$

12.8.

Because the solution contains chloride ions, which react with mercury(I) ions from the reference electrode.

12.9.

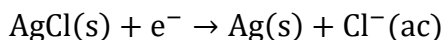
When Ag^+ ion is in excess



$$\Delta E = E_{\text{cat}} - E_{\text{anod}} = \left(E^\circ_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{1}{c(\text{Ag}^+)} \right) - E^\circ_{\text{Cal}}$$

$$\Delta E = E^\circ_{\text{Ag}^+/\text{Ag}} - E^\circ_{\text{Cal}} + \frac{R \cdot T}{F} \cdot \ln c(\text{Ag}^+)$$

When Cl^- is in excess



$$\Delta E = E_{\text{cat}} - E_{\text{anod}} = \left(E^\circ_{\text{AgCl}/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-) \right) - E^\circ_{\text{Cal}}$$

$$\Delta E = E^\circ_{\text{AgCl}/\text{Ag}} - E^\circ_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-)$$

12.10.

In mixtures 1-5, Ag^+ io is in excess. For example, in mixture 1

$$E^\circ_{\text{Ag}^+/\text{Ag}} = \Delta E + E^\circ_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln c(\text{Ag}^+)$$

$$E^{\circ}_{\text{Ag}^+/\text{Ag}} = 0,499 + 0,241 - \frac{8,3145 \cdot 298,15}{96485} \cdot \ln 0,100 = 0,799 \text{ V}$$

In mixtures 7-10, Cl^- ion is in excess. For example, in mixture 10

$$c(\text{Cl}^-) = \frac{c(\text{KCl}) \cdot V(\text{KCl}) - c(\text{AgNO}_3) \cdot V(\text{AgNO}_3)}{V(\text{KCl}) + V(\text{AgNO}_3)} = \frac{0,100 \cdot 90 - 0,100 \cdot 10}{90 + 10}$$

$$c(\text{Cl}^-) = 0,0800 \text{ mol} \cdot \text{L}^{-1}$$

$$E^{\circ}_{\text{AgCl}/\text{Ag}} = \Delta E + E^{\circ}_{\text{Cal}} + \frac{R \cdot T}{F} \cdot \ln c(\text{Cl}^-)$$

$$E^{\circ}_{\text{AgCl}/\text{Ag}} = 0,046 + 0,241 + \frac{8,3145 \cdot 298,15}{96485} \cdot \ln 0,08 = 0,222 \text{ V}$$

It is possible to build a linear regression with the data on the table.

12.11.

In mixture 6, the mixed amounts are stoichiometrically, then

$$[\text{Ag}^+] = [\text{Cl}^-] = \sqrt{K_{\text{ps}}}$$

$$E^{\circ}_{\text{Ag}^+/\text{Ag}} = \Delta E + E^{\circ}_{\text{Cal}} - \frac{R \cdot T}{F} \cdot \ln[\text{Ag}^+]$$

$$[\text{Ag}^+] = \exp\left(\frac{E^{\circ}_{\text{Ag}^+/\text{Ag}} - \Delta E - E^{\circ}_{\text{Cal}}}{-\frac{R \cdot T}{F}}\right) = \exp\left(\frac{0,799 - 0,269 - 0,241}{-\frac{8,3145 \cdot 298,15}{96485}}\right)$$

$$[\text{Ag}^+] = 1,30 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$K_{\text{ps}} = [\text{Ag}^+] \cdot [\text{Cl}^-] = (1,30 \cdot 10^{-5})^2 = 1,70 \cdot 10^{-10}$$

Alternatively, in the AgCl/Ag system, when $c(\text{Cl}^-) = 1 \text{ mol} \cdot \text{L}^{-1}$

$$E_{\text{Ag}^+/\text{Ag}} = E^{\circ}_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{1}{c(\text{Ag}^+)}$$

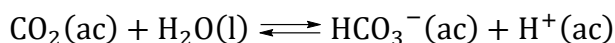
$$E^{\circ}_{\text{AgCl}/\text{Ag}} = E^{\circ}_{\text{Ag}^+/\text{Ag}} - \frac{R \cdot T}{F} \cdot \ln \frac{[\text{Cl}^-]}{K_{\text{ps}}}$$

$$K_{\text{ps}} = \exp\left(\frac{E^{\circ}_{\text{AgCl}/\text{Ag}} - E^{\circ}_{\text{Ag}^+/\text{Ag}}}{-\frac{R \cdot T}{F}}\right) \cdot [\text{Cl}^-] \exp\left(\frac{0,222 - 0,799}{-\frac{8,3145 \cdot 298,15}{96485}}\right) \cdot 1,00 = 1,76 \cdot 10^{-10}$$

Problem 13**13.1.**

$$c(\text{CO}_2) = 0,0752 \text{ M} \quad n(\text{CO}_2) = 0,0752 \text{ mol}$$

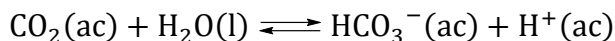
$$V = \frac{nRT}{p} = \frac{0,0752 \cdot 8,314 \cdot 273,15}{100} = 1,71 \text{ dm}^3$$

13.2.

$$[\text{H}^+] = [\text{HCO}_3^-(\text{ac})] = x \quad y \quad [\text{CO}_2] + [\text{HCO}_3^-] = 0,0752$$

$$K_a = 10^{-6,63} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]} = \frac{x^2}{0,0752 - x}$$

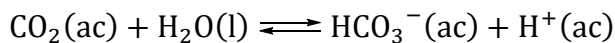
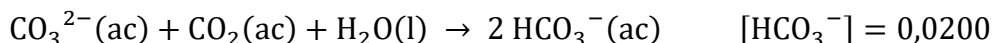
$$x = [\text{H}^+] = 0,000133 \quad y \quad [\text{CO}_2] = 0,0751$$

13.3.

$$[\text{CO}_2] = 0,0751 \quad y \quad [\text{HCO}_3^-] = 0,0100$$

$$K_a = 10^{-6,63} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]} = \frac{x \cdot 0,0100}{0,0751}$$

$$x = [\text{H}^+] = 1,76 \cdot 10^{-6}$$

13.4.

$$K_a = 10^{-6,63} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]} = \frac{x \cdot 0,0200}{0,0751}$$

$$x = [\text{H}^+] = 8,8 \cdot 10^{-7}$$

13.5.

$$0,0012 \text{ g}/100,1 \text{ g} \cdot \text{mol}^{-1} \text{ in } 100 \text{ cm}^3 \text{ of water}$$

$$[\text{Ca}^{2+}] = 1,2 \cdot 10^{-4} \quad c(\text{Ca}^{2+}) = 1,2 \cdot 10^{-4} \text{ mol} \cdot \text{dm}^{-3}$$

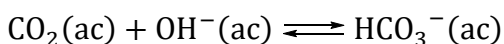
13.6.

$$K = \frac{[\text{Ca}^{2+}][\text{HCO}_3^-]^2}{[\text{CO}_2]} = 10^{-4,25} \quad \text{y} \quad 2[\text{Ca}^{2+}] = [\text{HCO}_3^-]$$

$$\frac{4[\text{Ca}^{2+}]^3}{0,0751} = 10^{-4,25} \quad [\text{Ca}^{2+}] = 1,02 \cdot 10^{-2} \quad c(\text{Ca}^{2+}) = 1,02 \cdot 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$$

13.7.

$$c(\text{Ca}(\text{OH})_2) = 0,015 \text{ mol} \cdot \text{dm}^{-3}$$



All the hydroxide is consumed ($K = 10^{7,37}$).

From item 13.6, it is known that the maximum concentration of calcium ions is lower than this value, therefore, CaCO_3 precipitates.

$$[\text{Ca}^{2+}] = 1,02 \cdot 10^{-2} \quad c(\text{Ca}^{2+}) = 1,02 \cdot 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$$

13.8.

$$c(\text{Ca}(\text{OH})_2) = 0,0075 \text{ mol} \cdot \text{dm}^{-3}$$

From item 13.6, it is known that the maximum concentration of calcium ions is higher than this value, therefore, CaCO_3 does not precipitate.

$$[\text{Ca}^{2+}] = 0,75 \cdot 10^{-2} \quad c(\text{Ca}^{2+}) = 0,75 \cdot 10^{-2} \text{ mol} \cdot \text{dm}^{-3}$$

13.9.

$$K = \frac{[\text{Ca}^{2+}][\text{HCO}_3^-]^2}{[\text{CO}_2]} = \frac{[\text{Ca}^{2+}][\text{HCO}_3^-]^2}{[\text{CO}_2]} \cdot \frac{[\text{CO}_3^{2-}][\text{H}^+]}{[\text{CO}_3^{2-}][\text{H}^+]} = \frac{K_{\text{ps}} \cdot K_{\text{a1}}}{K_{\text{a2}}}$$

$$K_{\text{ps}} = 10^{-8,26}$$

Problem 14**14.1.**

$$\Delta H_r^\circ = \Delta H_f^\circ[\text{I}_2(\text{g})] - \Delta H_f^\circ[\text{I}_2(\text{s})] = 62,3 - 0 = 62,3 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta S^\circ_r = S^\circ[\text{I}_2(\text{g})] - S^\circ[\text{I}_2(\text{s})] = 260,7 - 116,7 = 144 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$$

$$\Delta G^\circ_r = -RT \ln p_r(\text{I}_2) = -8,314 \cdot 298 \cdot \ln 1,013 = -32,0 \text{ J} \cdot \text{mol}^{-1}$$

$$T = \frac{\Delta G^\circ_r - \Delta H^\circ_r}{-\Delta S^\circ_r} = \frac{-32 - 62300}{-144} = 433 \text{ K} = 160^\circ\text{C}$$

14.2.

$$\Delta G^\circ_r = \Delta G^\circ_f[\text{I}_2(\text{CH}_2\text{Cl}_2)] - \Delta G^\circ_f[\text{I}_2(\text{ac})]$$

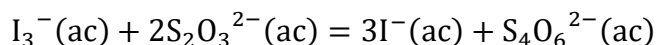
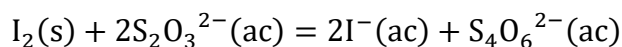
$$\Delta G^\circ_r = 4,00 - 16,4 = -12,4 \text{ kJ} \cdot \text{mol}^{-1}$$

$$K_2 = \exp\left(\frac{\Delta G^\circ_r}{-RT}\right) = \exp\left(\frac{-12400}{-8,314 \cdot 298}\right) = 149$$

14.3.

Iodine is an apolar substance like most of organic solvents; while water is a polar substance.

14.4.



Oxidation numbers of sulfur are 4+ and 0. Answer 2+ is accepted.

14.5.

$$[\text{I}_2(\text{ac})] = \frac{[\text{I}_2(\text{CH}_2\text{Cl}_2)]}{K_2} = \frac{1,2 \cdot 10^{-2}}{149} = 8,05 \cdot 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

$$n[\text{I}_2(\text{ac})] + n[\text{I}_3^-(\text{ac})] = \frac{1}{2} \cdot n(\text{S}_2\text{O}_3^{2-}) = \frac{1}{2} \cdot 0,0100 \cdot 3,10 = 1,55 \cdot 10^{-2} \text{ mmol}$$

$$[\text{I}_2(\text{ac})] + [\text{I}_3^-(\text{ac})] = \frac{1,55 \cdot 10^{-2}}{25,0} = 6,20 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{I}_3^-(\text{ac})] = 6,20 \cdot 10^{-4} - 8,05 \cdot 10^{-5} = 5,40 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{I}^-(\text{ac})] = 0,0100 - [\text{I}_3^-(\text{ac})] = 0,0100 - 5,40 \cdot 10^{-4} = 9,46 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1}$$

$$K_3 = \frac{[\text{I}_3^-(\text{ac})]}{[\text{I}_2(\text{ac})][\text{I}^-(\text{ac})]} = \frac{5,40 \cdot 10^{-4}}{8,05 \cdot 10^{-5} \cdot 9,46 \cdot 10^{-3}} = 709$$

14.6.

For equilibrium number (3) it is true that

$$\Delta E_r^\circ = E^\circ(\text{I}_2/\text{I}^-) - E^\circ(\text{I}_3^-/\text{I}^-) = 0,621 - 0,536 = 0,085 \text{ V}$$

$$K_3 = \exp\left(\frac{-zF\Delta E_r^\circ}{-RT}\right) = \exp\left(\frac{-2 \cdot 96485 \cdot 0,085}{-8,314 \cdot 298}\right) = 750$$

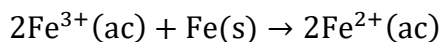
Problem 15**15.1.**

Iron(III) is more oxidant in acid media.

Metallic iron is more reductant in basic media.

15.2.

No. For none of the species the reduction potential is higher than the oxidation potential.

15.3.

$$\Delta E^\circ = E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) - E^\circ(\text{Fe}^{2+}/\text{Fe}) = 0,77 - (-0,44) = 1,21 \text{ V}$$

No, the potential difference is positive

15.4.

$$3E^\circ(\text{Fe}^{3+}/\text{Fe}) = E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) + 2E^\circ(\text{Fe}^{2+}/\text{Fe})$$

$$E^\circ(\text{Fe}^{3+}/\text{Fe}) = \frac{0,77 + 2 \cdot (-0,44)}{3} = -0,037 \text{ V}$$

15.5.

$$p(\text{O}_2) = \chi(\text{O}_2) \cdot p(\text{total}) = 0,21 \cdot 1,01 = 0,212 \text{ bar}$$

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-7,0} = 1,00 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1}$$

$$E = E^\circ - \frac{RT}{4F} \ln\left(\frac{1}{p_r(\text{O}_2)[\text{H}_3\text{O}^+]^4}\right)$$

$$E = 1,23 - \frac{8,3145 \cdot 298,15}{4 \cdot 96485} \ln \left(\frac{1}{0,212 \cdot [10^{-7}]^4} \right) = 0,81 \text{ V}$$

15.6.

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-6,5} = 3,16 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1}$$

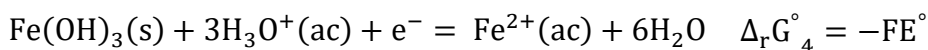
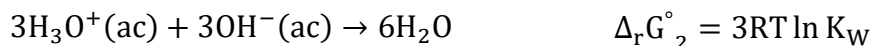
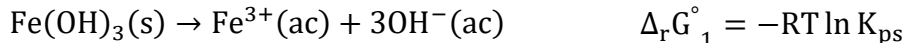
$$[\text{OH}^-] = \frac{K_W}{[\text{H}_3\text{O}^+]} = \frac{1,00 \cdot 10^{-14}}{3,16 \cdot 10^{-7}} = 3,16 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Fe}^{3+}] = \frac{K_{ps}}{[\text{OH}^-]^3} = \frac{2,79 \cdot 10^{-39}}{[3,16 \cdot 10^{-8}]^3} = 8,84 \cdot 10^{-17} \text{ mol} \cdot \text{L}^{-1}$$



$$E = E^\circ - \frac{RT}{F} \ln \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) \rightarrow [\text{Fe}^{2+}] = [\text{Fe}^{3+}] \cdot \exp \left(\frac{F(E - E^\circ)}{-RT} \right)$$

$$[\text{Fe}^{2+}] = 8,84 \cdot 10^{-17} \cdot \exp \left(\frac{96485 \cdot (0,65 - 0,77)}{-8,3145 \cdot 298,15} \right) = 9,44 \cdot 10^{-15} \text{ mol} \cdot \text{L}^{-1}$$

15.7.

$$\Delta_r G^\circ_4 = \Delta_r G^\circ_1 + \Delta_r G^\circ_2 + \Delta_r G^\circ_3$$

$$-FE^\circ = -RT \ln K_{ps} + 3RT \ln K_W - FE^\circ(\text{Fe}^{3+}/\text{Fe}^{2+})$$

$$E^\circ = - \frac{-8,314 \cdot 298,15 \cdot \ln 2,79 \cdot 10^{-39} + 3 \cdot 8,314 \cdot 298,15 \cdot \ln 1,00 \cdot 10^{-14} - 96485 \cdot 0,77}{96485}$$

$$E^\circ = 0,974 \text{ V}$$

15.8.

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-6,0} = 1,00 \cdot 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{OH}^-] = \frac{K_W}{[\text{H}_3\text{O}^+]} = \frac{1,00 \cdot 10^{-14}}{1,00 \cdot 10^{-6}} = 1,00 \cdot 10^{-8} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Fe}^{3+}] = \frac{K_{\text{ps}}}{[\text{OH}^-]^3} = \frac{2,79 \cdot 10^{-39}}{[1,00 \cdot 10^{-8}]^3} = 2,79 \cdot 10^{-15} \text{ mol} \cdot \text{L}^{-1}$$



$$E = E^\circ - \frac{RT}{F} \ln \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) \rightarrow [\text{Fe}^{2+}] = [\text{Fe}^{3+}] \cdot \exp \left(\frac{F(E - E^\circ)}{-RT} \right)$$

$$[\text{Fe}^{2+}] = 2,79 \cdot 10^{-15} \cdot \exp \left(\frac{96485 \cdot (0,15 - 0,77)}{-8,3145 \cdot 298,15} \right) = 1,88 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

The reduction half equation from the previous item can also be employed for the calculation.

Problem 16

16.1.

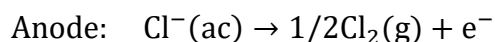
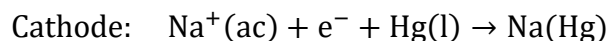


$$\Delta E_r^\circ = E^\circ_{\text{cat}} - E^\circ_{\text{anod}} = E^\circ_{\text{MnO}_2/\text{Mn}^{2+}} - E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1,23 - 1,358 = -0,128 \text{ V}$$

$$\Delta G_r^\circ = -z \cdot F \cdot \Delta E_r^\circ = -2 \cdot 96485 \cdot (-0,128) = 24700 \text{ J} \cdot \text{mol}^{-1}$$

$$K^\circ = \exp \left(-\frac{\Delta G_r^\circ}{R \cdot T} \right) = \exp \left(-\frac{24700}{8,314 \cdot 298,15} \right) = 4,70 \cdot 10^{-5}$$

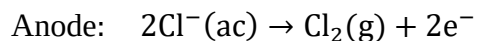
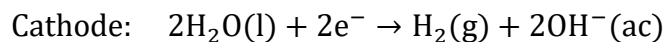
16.2.



16.3.

Mercury is required and this is a very toxic element.

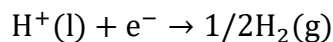
16.4.



16.5.

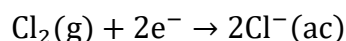
$$M(\text{NaCl}) = A_r(\text{Na}) + A_r(\text{Cl}) = 22,99 + 35,45 = 58,44 \text{ g} \cdot \text{mol}^{-1}$$

$$c(\text{NaCl}) = \frac{\omega_{\%} \cdot d \cdot 10}{M(\text{NaCl})} = \frac{26,0 \cdot 1,20 \cdot 10}{58,44} = 5,34 \text{ mol} \cdot \text{L}^{-1}$$

16.6.

$$E_{\text{H}^+/\text{H}_2} = E^\circ_{\text{H}^+/\text{H}_2} - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{p_r(\text{H}_2)}{c(\text{H}^+)}$$

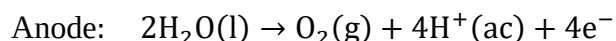
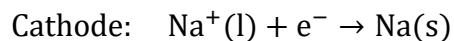
$$E_{\text{H}^+/\text{H}_2} = 0,00 - \frac{8,3145 \cdot 298}{2 \cdot 96485} \cdot \ln \frac{1}{1 \cdot 10^{-7}} = -0,414 \text{ V}$$



$$E_{\text{Cl}_2/\text{Cl}^-} = E^\circ_{\text{Cl}_2/\text{Cl}^-} - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c^2(\text{Cl}^-)}{p_r(\text{Cl}_2)}$$

$$E_{\text{Cl}_2/\text{Cl}^-} = 1,358 - \frac{8,3145 \cdot 298}{2 \cdot 96485} \cdot \ln \frac{(5,34)^2}{1} = 1,315 \text{ V}$$

$$\Delta E_{\min} = |E_{\text{cat}} - E_{\text{anod}}| = |E_{\text{H}^+/\text{H}_2} - E_{\text{Cl}_2/\text{Cl}^-}| = |-0,414 - 1,315| = 1,729 \text{ V}$$

16.7.

The cathodic half reaction does not occur because it is thermodynamically not favored.

The anodic half reaction does not occur because it is kinetically not favored.

16.8.

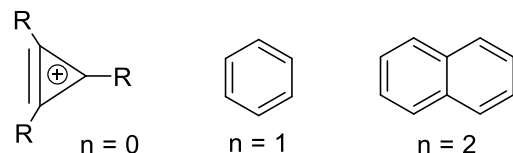
$$n(\text{e}^-) = 2 \cdot n(\text{Cl}_2) = 2 \cdot \frac{m(\text{Cl}_2)}{M(\text{Cl}_2)} = 2 \cdot \frac{1,42 \cdot 1000}{70,9} = 40,1 \text{ mol}$$

$$q = F \cdot n(\text{e}^-) = 96485 \cdot 40,1 = 3,86 \cdot 10^6 \text{ C}$$

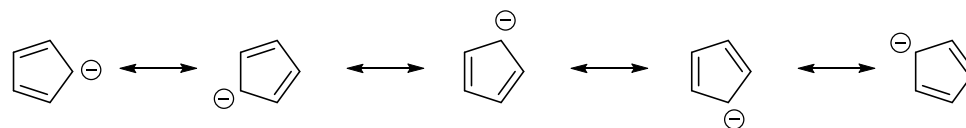
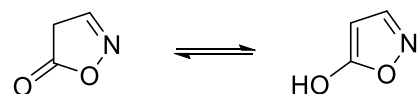
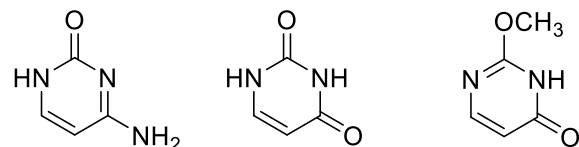
$$\text{Energy} = q \cdot \Delta E_{\min} = 3,86 \cdot 10^6 \cdot 1,729 = 6,67 \cdot 10^6 \text{ J} = 1,85 \text{ kW} \cdot \text{h}$$

16.9.

Cl_2 and ClO_2^- disproportionate.

16.10.**Problem 17****17.1.****17.2.**

The conjugate base of cyclopentadiene is aromatic.

**17.3.****17.4.**

The first is cytosine, the second is uracil. The essential role of those tautomers is to engage the hydrogen bonds between the nucleic acid chains. Non-aromatic tautomers form a larger number of hydrogen bonds between the DNA base pairs. Other examples are guanine and thymine. Adenine exists in the aromatic form.

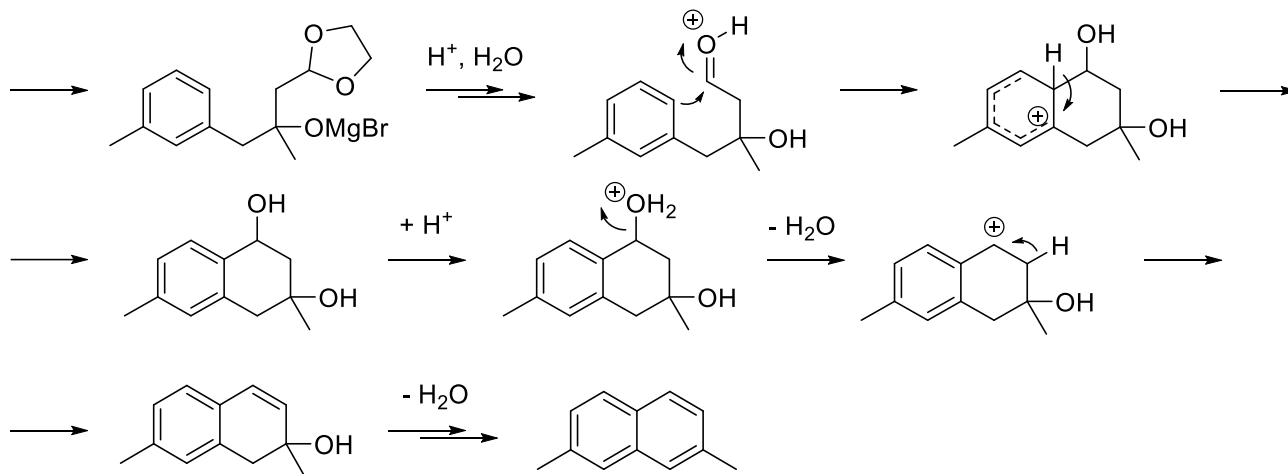
Problem 18

18.1.

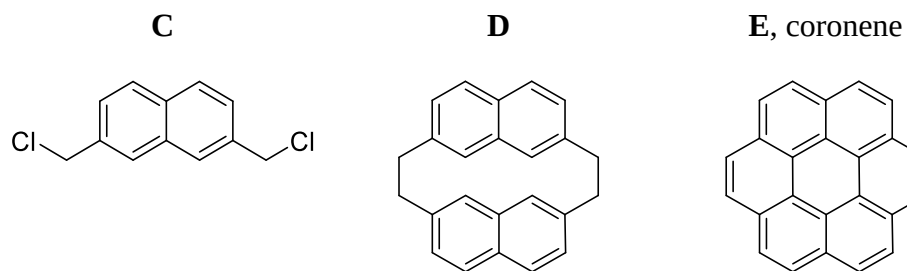
i) 1. NBS (1 equiv.), 2. Mg / éter (I2 cat.)

ii) HOCH₂CH₂OH, pTsOH

18.2.

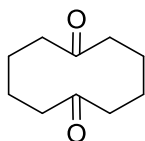
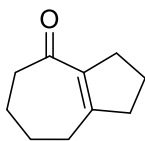
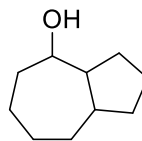
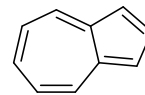
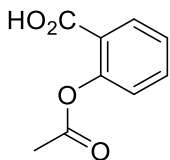
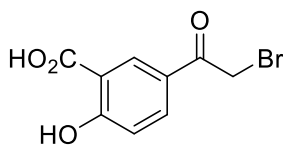
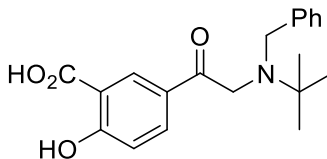
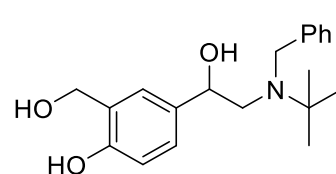


18.3.

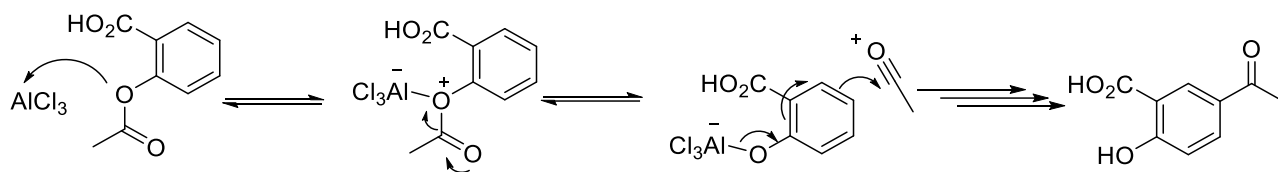
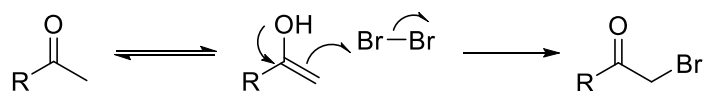


Problem 19



E**F****G****H****Problem 20****20.1.****A****B****C****D****20.2.**

Medicine is aspirine.

**20.3.****20.4.**

To avoid over alkylation of the amine.

Annex 1. Results of the Cuban National Chemistry Contest 2018²⁰

Results by provinces

Province	O	P	B	Total	PSN	Points	Place
CM	4	4	1	9	9	45	1
SS	3	4	2	9	7	43	2
MT	2	4	3	9	7	37	3
LT	0	6	4	10	6	34	4
LH	1	4	4	9	2	25	5
SC	1	0	5	6	1	12	6
VC	0	1	4	5	1	9	7
AR	1	0	2	3	1	9	7
CF	0	0	4	4	0	4	9
GR	0	0	4	4	0	4	9
GT	0	0	4	4	0	4	9
PR	0	0	3	3	0	3	12
IJ	0	0	1	1	0	1	13
MY	0	0	0	0	0	0	14
CA	0	0	0	0	0	0	14
HO	0	0	0	0	0	0	14
UH	0	0	0	0	0	0	14
UVC	0	0	0	0	0	0	14

Qualification tribunal

MSc. Luis Enrique Guerra Castaño	LH	IPVCE Vladimir Ilich Lenin
MSc. Julio Rodríguez Benítez	CM	IPVCE Máximo Gómez Baéz
MSc. Agustín Plasencia Calero	SS	IPVCE Eusebio Olivera Labra
MSc. Juan Francisco Olivera	PR	IPVCE Federico Engels
MSc. Asterio Marrero Machado	GR	IPVCE Silberto Álvarez Aroche
MSc. Rafael León Pérez	LH	IPVCE Vladimir Ilich Lenin
BSc. Jorge Alberto Arias Rodríguez	SC	IPVCE Antonio Maceo Grajales
MSc. José María Fernández García	AT	Universidad Tecnológica de La Habana

Authors of the problems

MSc. Gerardo M. Ojeda Carralero
BSc. Asiel Mena Jiménez

Revision

MSc. Leonardo González Ceballos
Javier E. Alfonso Ramos

²⁰ Symbology:

O: Gold Medal (5 points)

P: Silver Medal (3 points)

B: Bronze Medal (1 point)

PSN: National Pre-selection (2 points)

Tenth grade

Héctor David Couso Hernández	SS	IPVCE Eusebio Olivera Rodríguez	84,50	O
Favio Darío Quevedo Sedeño	CM	IPVCE Máximo Gómez Báez	84,50	O
Sheyla María Peña González	CM	IPVCE Máximo Gómez Báez	84,00	O
Alessia Pérez Daniel	MT	IPVCE Carlos Marx Pressburg	80,50	O
Sahiary Martínez Paz	CM	IPVCE Máximo Gómez Báez	68,00	P
Lázaro Arturo Ichaso Pérez	SS	IPVCE Eusebio Olivera Rodríguez	63,10	P
Mariana Gamboa Céspedes	LT	IPVCE Luis Urquiza Jorge	62,76	P
Hubert César Cabrera Betancourt	MT	IPVCE Carlos Marx Pressburg	62,08	P
Yanitza Yisay Céspedes Echavarría	LT	IPVCE Luis Urquiza Jorge	54,08	P
Isabella Maciel Martínez Núñez	LT	IPVCE Luis Urquiza Jorge	54,08	P
Johan Ernesto Méndez Batista	LH	IPVCE Vladimir Ilich Lenin	51,03	P
Christian José Saavedra Díaz	SS	IPVCE Eusebio Olivera Rodríguez	50,53	P
Brayan González Falcón	MT	IPVCE Carlos Marx Pressburg	45,23	B
Ricardo Alejandro Duarte Griñán	GT	IPVCE José Maceo Grajales	43,19	B
Laura Rodríguez Martínez	SC	IPU Rafael María de Mendive	42,81	B
David Alejandro Pérez León	AR	IPVCE Mártires de Humboldt 7	41,37	B
Camila Viviana Rosales Olivera	GR	IPVCE Silberto Álvarez Aroche	41,10	B
Maylis Herrera Pacheco	VC	IPVCE Ernesto Che Guevara	40,96	B
Glenda Bauta Castañeda	LH	IPVCE Vladimir Ilich Lenin	39,33	B
Liovanys Damas Sariol	SS	IPVCE Eusebio Olivera Rodríguez	36,00	B
Carmen Gloria Dean Galán	SC	IPU Rafael María de Mendive	34,73	B
Jeniffer Bravo Martín	LH	IPVCE Vladimir Ilich Lenin	34,62	B
Robert Jordán Martínez	SC	IPU Rafael María de Mendive	33,23	B
Magdiel Enrique Martínez Batista	GR	IPVCE Silberto Álvarez Aroche	32,62	B
Liana María Correoso Heredia	SC	IPU Rafael María de Mendive	32,07	B
Karen Rachel Corrales Cruz	GR	IPVCE Silberto Álvarez Aroche	31,84	B
Claudia Caridad Cuéllar Cabrera	CF	IPVCE Carlos Roloff Mialofsky	31,50	B
Sergio David Roche Díaz	LH	IPVCE Vladimir Ilich Lenin	30,62	B
Martha Laura López Rubio	CF	IPVCE Carlos Roloff Mialofsky	30,30	B
William Benítez Rodríguez	MT	IPVCE Carlos Marx Pressburg	27,35	B
Lidice Fabiana Pérez Palma	GR	IPVCE Silberto Álvarez Aroche	27,19	B

Eleventh grade

Davier Sánchez Bello	MT	IPVCE Carlos Marx Pressburg	87,06	O
Rey Michel Rodríguez Navarro	CM	IPVCE Máximo Gómez Báez	84,94	O
Jocdan Lismar López Mantecón	CM	IPVCE Máximo Gómez Báez	82,92	O
Laura Barreto González	AR	IPVCE Mártires de Humboldt 7	74,52	O
David Linares Ramírez	VC	IPVCE Ernesto Che Guevara	72,41	P
Rafael Alejandro Perurena García	SS	IPVCE Eusebio Olivera Rodríguez	72,21	P
Massiel B. Fuentes Vázquez	MT	IPVCE Carlos Marx Pressburg	71,62	P
José Manuel Cabrera Pereira	SS	IPVCE Eusebio Olivera Rodríguez	71,55	P
Samuel Báez Lau	LH	IPVCE Vladimir Ilich Lenin	70,33	P
Eduardo Addiel Landrove Escalona	LT	IPVCE Luis Urquiza Jorge	68,78	P
Rachel Yina Báez González	LT	IPVCE Luis Urquiza Jorge	67,78	P
Osniel García Orihuela	MT	IPVCE Carlos Marx Pressburg	67,17	P
José Ángel Delgado Morejón	LH	IPVCE Vladimir Ilich Lenin	64,17	P
Camilo Mas Jacobs	LH	IPVCE Vladimir Ilich Lenin	64,14	P
Lisvany Martín Rodríguez	VC	IPVCE Ernesto Che Guevara	62,19	B
Marcos Antonio Fernández Roquero	CM	IPVCE Máximo Gómez Báez	60,43	B
Maday Sánchez Oramas	MT	IPVCE Carlos Marx Pressburg	59,00	B
Ernesto Alejandro Fajardo Díaz	AR	IPVCE Mártires de Humboldt 7	57,19	B
Laura Gil Espinosa	LH	IPVCE Vladimir Ilich Lenin	55,43	B
Anier Alcides Peña Omar	GT	IPVCE José Maceo Grajales	52,47	B
Ana Elia Miranda Castañeda	PR	IPVCE Federico Engels	52,34	B
Beatriz Reyes Manals	SC	IPU Rafael María de Mendive	52,16	B
Laura Maday De la Paz Cordero	LT	IPVCE Luis Urquiza Jorge	48,99	B
Amanda Ivet Rivero Arbolay	VC	IPVCE Ernesto Che Guevara	48,25	B
Claudia Ruiz Nogueira	PR	IPVCE Federico Engels	48,09	B

Twelfth grade

Sergio Modesto Reyes Díaz	LH	IPVCE Vladimir Ilich Lenin	90,15	O
Isaac Enrique Peñuela Puente	SC	IPVCE Antonio Maceo Grajales	85,00	O
Adolfo Alejandro Cruz Cruz	SS	IPVCE Eusebio Olivera Rodríguez	83,35	O
Carlos Rafael Castillo Hernández	SS	IPVCE Eusebio Olivera Rodríguez	82,65	O
Adiel Martínez Rodríguez	MT	IPVCE Carlos Marx Pressburg	79,35	P
Josué Ponce de León Cabrera	CM	IPVCE Máximo Gómez Báez	78,86	P
Lilisbeth De La Caridad Bénitez Rojas	LT	IPVCE Luis Urquiza Jorge	78,73	P
Maricary Sifontes Niebla	SS	IPVCE Eusebio Olivera Rodríguez	74,35	P
Ariel Ernesto Martínez Suffos	CM	IPVCE Máximo Gómez Báez	73,93	P
Ariel Alejandro Castro Sifonte	CM	IPVCE Máximo Gómez Báez	73,85	P
Denys Daniel Núñez Pupo	LT	IPVCE Luis Urquiza Jorge	62,10	B
Lilianne Gil Batista	LT	IPVCE Luis Urquiza Jorge	59,71	B
Sara María Collejo Fernández	LT	IPVCE Luis Urquiza Jorge	55,08	B
Manuel Alejandro Santana Hernández	VC	IPVCE Ernesto Che Guevara	46,46	B
Rolando Pérez Heredia	GT	IPVCE José Maceo Grajales	42,89	B
Griela María Águila Álvarez	PR	IPVCE Federico Engels	40,83	B
Jorge Ernesto Claro Díaz	GT	IPVCE José Maceo Grajales	31,33	B
Thalía Yera Martínez	CF	IPVCE Carlos Roloff Mialofsky	29,90	B
Ismanuel Basulto Díaz	IJ	IPU Rodolfo Carballosa	29,76	B
Noel Jesús Niebla Gómez	CF	IPVCE Carlos Roloff Mialofsky	28,83	B

Annex 2. Results obtained in the international chemistry olympiads in 2017 and 2018

49th International Chemistry Olympiad

Celebrated on July 6 to 15, 2017 in Nakhon Pathom, Thailand. The exams took place in the University of Mahidol. 76 countries around the world participated. The members of the Cuban delegation were:

MSc. Luis Enrique Guerra Castaño (Mentor)
IPVCE “Vladimir Ilich Lenin”, La Habana

José Rafael Rodríguez Rodríguez (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 12th grade

Bronze Medal

XXII Ibero-American Chemistry Olympiad

Celebrated on October 8 to 15, 2017 in Lima, Peru. The exams took place in the Pontificia Catholic University of Peru. 17 countries from Latin America and the Iberic Peninsula participated in the competition. The members of the Cuban delegation were:

MSc. Rolando Alfonso Valdés (Mentor)
IPVCE “Ernesto Che Guevara”, Villa Clara

Carlos Rafael Castillo Hernández (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 11th grade

Gold Medal



Luis Guerra and José Rafael in the 49th International Chemistry Olympiad.

XII Central-American and X Caribbean Chemistry Olympiad

Celebrated on July 8 to 14, 2018 in Guatemala City, Guatemala. The examinations took place in the University of San Carlos of Guatemala. 7 countries from Central-American and the Caribbean participated. The Cuban delegation was:

MSc. Rolando Alfonso Valdés (Mentor)
IPVCE “Ernesto Che Guevara”, Villa Clara

Adolfo Alejandro Cruz Cruz (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 12th grade

Medalla de Oro

Osniel García Orihuela (Student)
IPVCE “Carlos Marx Pressburg”, Matanzas, 11th grade

Medalla de Plata

Adiel Martínez Rodríguez (Student)
IPVCE “Carlos Marx Pressburg”, Matanzas, 12th grade

Medalla de Plata

Jocdan Lismar López Mantecón (Student)
IPVCE “Máximo Gómez Báez”, Camagüey, 11th grade

Medalla de Bronce

XXIII Iberoamerican Chemistry Olympiad

Celebrated on September 21 to 28, 2018 in San Salvador, El Salvador. The examinations took part at the University of El Salvador. 17 countries from Latin-America and the Iberic Peninsula participated. The Cuban delegation was:

MSc. Orestes Landrove Ramírez (Mentor)
IPVCE “Luis Urquiza Jorge”, Las Tunas

Rey Michel Rodríguez Navarro (Student)
IPVCE “Máximo Gómez Báez”, Camagüey, 11th grade

Silver Medal

Carlos Rafael Castillo Hernández (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 12th grade

Silver Medal

Sergio Modesto Reyes Díaz (Student)
IPVCE “Vladimir Ilich Lenin”, La Habana, 12th grade

Silver Medal

Isaac Enrique Peñuela Puente (Student)
IPVCE “Antonio Maceo Grajales”, Santiago de Cuba, 12th grade

Bronze Medal



Participants in the XII Central-American and X Caribbean Chemistry Olympiad (from left to right): Adiel, Adolfo Alejandro, Rolando Alfonso, Jocdan Lismar and Osniel.



Participants in the XXIII Ibero-American Chemistry Olympiad (from left to right): Rey Michel, Sergio Modesto, Carlos Rafael, Isaac Enrique and Orestes.

Annex 3. Topics of Organic Chemistry for the National Contest

Fundamentals of structure, bonding and reactivity applied to the organic molecules

Nomenclature of the main families of organic molecules.

Stereochemistry. Elements of molecular chirality, asymmetric center, plane and symmetry axes; atoms that may be asymmetric (C, S, N, P); chirality of a molecule, chirality of allenes and biphenyls; absolute stereochemistry and relative stereochemistry; notation of Cahn-Ingold-Prelog; use of tridimensional representations; E and Z notation for alkenes; kind of stereoisomers, physical and chemical properties, optical activity; racemization and racemic mixture; meso, erythro and threo stereoisomers.

Conformational analysis. Conformations of alkanes and cycloalkanes; torsion tension and sterical hindrance; eclipsed, anti and gauche conformations; envelope conformation of cyclopentane; chair and boat conformations of cyclohexane, axial and equatorial positions; see Newman's and Caballero's projections; cis and trans fusions in cyclic compounds and in fused bicycles; Bredt's rule for bridged bicycles.

Intermediaries in organic reactions: carbocations, carbanions, radicals and carbenes; formation; relative stability.

Stereoelectronic effects. Resonance, conjugation and hyperconjugation; inductive effect; mesomeric effect; aromaticity and antiaromaticity, Hückel's rule; electron rich π rings and electron deficient π rings; neighboring group effect (anchimeric assistance).

Reactivity. Nucleophiles and electrophiles; leaving groups; acidity and basicity of organic molecules, determinant factors; comparison between S_N1 , S_N2 , E1 and E2, factors that favor a particular mechanism (electronic effects, sterical hindrance, strength of the nucleophile/base, solvent and temperature, kinetics, transposition); quantitative calculation of the regioselectivity of the halogenation, difference between reactivity and selectivity, Hammond postulate; Markovnikov's rule for the electrophilic addition of alkenes and alkynes, peroxide effect or Kharasch-Mayo effect in the addition of HBr to alkenes; Pearson's theory of reactivity or hard and soft acid base theory, Michael's addition vs 1,2 addition, S_N2 and S_N2' reactions in allylic systems; causes of the distinct reactivity of alkenes, alkynes, dienes and aromatic derivatives; kinetic control and thermodynamic control in the reactions of conjugated dienes (1,2 and 1,4 products) and in the Diels-Alder's reaction

(endo and exo products); relative stability of the σ complex in S_NAr and S_EAr reactions, activants and deactivant groups, orientation of the reactions in substituted derivatives of benzene, furane, pyrrole, thiophane, pyridine, naphthalene, anthracene and phenanthrene; regioselectivity and stereoselectivity of the chemical reactions; relative reactivity of the carbonylic compounds towards nucleophiles, influence of steric and electronic effects and of the leaving group.

Radical reactions

Alkane halogenation; Kolbe's synthesis of alkanes; allylic and benzylic halogenation; Hunsdiecker's reaction; alkane pyrolysis and combustion reaction.

Organometallic reagents

Oxidative addition of Mg or Li in alkyl, aryl and vinyl halides; alkyne deprotonation; reactions with carbon dioxide, formaldehyde, aldehyde and ketones; transmetalation reactions to form other organometallic reagents (Cu, Zn, Cd); coupling reactions between organometallic reagents and halides (Corey-House's synthesis, Wurtz's reaction, Ullmann's reaction).

Nucleophilic substitution in the saturated carbon

S_N1 and S_N2 reactions of halides, sulfonates (tosylates, mesylates, triflates) and similar; reactions with water, silver salts, thiols, alcohols, phosphines, amines, azides, cyanides, alkynes, Gilman's reagents, phthalimides (Gabriel's synthesis), phenols (Williamson's synthesis) and thiophenols, halides (Finkelstein's reaction), (thio)carboxylic acids; reactions of alcohols with hydrazides; exhaustive methylation with Me_2SO_4 and MeI ; methylation of phenols and carboxylic acids with diazomethane; reaction of alcohols with phosphorous halides; deprotection of phenolic ethers with BBr_3 ; epoxide opening in basic and acid media with water, alcohols, amines, cyanides, hydrides, azide; Ritter's reaction; epoxide formation from halohydrins; reaction of ethers with hydrogen halides and water in acid media.

Elimination reaction

$E1$ and $E2$ reaction with different kind of bases and leaving groups; Hofmann's and Zaitsev's rules; Hofmann's elimination; Cope's elimination; eliminations in carbocations; dehydration of alcohols; alkyne formation from vicinal dihalides and vinyl halides, regioselectivity; conjugate elimination ($E1cB$) in the products of aldolic addition and similar.

Electrophilic addition to alkenes and alkynes

Reaction of alkenes with halogens and similar (IBr, ClSR), peroxides (Prilezhaev's reaction), hydrogen halide, N-halosuccinimides, sulfuric acid, boranes; Simmons-Smith's reaction; ozonolysis (oxidative, reductive and neutral), halohydrins formation and similar; oxy(alkoxy)mercuration demercuration; electrophilic addition to dienes at low and high temperature; hydration of alkenes and alkynes; polymerization of acetylene.

Addition to electrophilic alkenes

Reaction of conjugated double bonds with electronwithdrawing groups (amides, esters, ketones, aldehydes, acyl chlorides, nitro...) with cyanhydric acid, Grignard's reagent, Gilman's reagent, organolithiums, ammonia, amines, sodium borohydride, lithium and aluminum hydride, thiols, alcohols, phenols, thiophenols, carboxylic acid salts, phosphines, peroxides, hydrogen halides, acetylides and enolates (Michael's reaction).

Nucleophilic addition to the carbonyl group

Formation of cyanohydrins; reaction of aldehydes and ketones with hydrides, organometallic reagents, water, alcohols and bisulfite.

Nucleophilic substitution in the carbonyl group with loss of an oxygen atom

Formation of linear and cyclic acetals and ketals; formation of 1,3-dithianes; reactions of aldehydes and ketones with amines, hydroxylamine, hydrazine, (2,4-dinitro)phenylhydrazine and semicarbazide; reaction with POCl₃ and PCl₅; reductive amination; Strecker's synthesis; Wittig's reaction.

Nucleophilic substitution at the carbonyl group

Relative reactivity of acyl chlorides, anhydrides, esters, amides and carboxylic acids; reactions with amines, thiols, alcohols, water and carboxylic acids in acid or basic media; reaction of carboxylic acids with SOCl₂, PCl₅ and oxalyl chlorides; partial and total hydrolysis of nitriles; reaction with organolithiums, Grignard's and Gilman's reagent; reduction with lithium aluminum hydride and diborane; formation of sulfonamides and tosylates; addition of amines and alcohols to isocyanates; reaction of diazomethane with acid halides.

Reactions of enols, enolates and similar

Alkylation and acylation of enamides; formation of enols and enolates of esters, ketones, aldehydes, acid halides, nitriles, sulfones, sulfoxides and nitro compounds; reaction of enols and enolates with alkyl halides, acyl halides, sulfonic acid esters, halogens and their equivalents (NIS), epoxides, nitrous acid; haloform reaction; reactions of substitution of 1,3-dithianes; aldolic addition and condensation; Claisen's and Dieckmann's reaction; malonic synthesis, decarboxylation of malonic acids; acetoacetic synthesis, decarboxylation of α -ketoacids and of α -ketoesters (Krapcho's reaction); Hell-Volhard-Zelinsky's reaction; analogues of aldolic and Claisen's reactions in compounds with active hydrogens (modifications of Knoevenagel, Doebner, Perkin, nitroaldolic or Henry, Reformatsky, Stobbe, Mannich, Darzens).

Electrophilic aromatic substitution

Halogenation, nitration, sulfonation of benzene and their derivatives; Friedel-Crafts's alkylation and acylation; Kolbé-Schmidt's carbonatation; reactions of Gattermann and Gattermann-Koch; Reimer-Tiemann's reaction; Hoesch's reaction; chloromethylation; nitrosation.

Nucleophilic aromatic substitution

Reaction of deactivated halobenzenes with amines, alcohols, thiols, water, hydrazine, cyanides, phenols, thiophenols and azide.

Cycloaddition reactions

Diels-Alder's reaction; alkene hydroxylation of OsO₄; ozonolysis; insertion of carbenes to alkenes, Simmons-Smith's reaction.

Rearrangements and fragmentations

Picolinic and benzylic acid rearrangements; reactions of Baeyer-Villiger and Dakin; migrations in carbocations; Beckmann's transposition; Hofmann's transposition; Tischenko's transposition.

Oxidations and reductions

Reactions of reduction with LiAlH₄, NaBH₄, H₂/catalyzer, B₂H₆, N₂H₂, NaBH₃CN, DIBALH, metal/NH₃(l); reduction of Birch, Zn(Hg)/HCl (reduction of Clemmensen), Sn/HCl, NH₂NH₂/base (reaction of Wolf-Kirchner), Na/alcohol; obtention of pinacols.

Oxidation reactions with KMnO_4 , HIO_4 , H_2CrO_4 , PCC, PDC, H_2O_2 , AMCPB, O_3 , OsO_4 . Cannizaro's reaction; Meerwein-Ponndorf-Verley's reduction and Oppenauer's oxidation; allylic oxidation with CrO_3 and SeO_2 ; Riley's oxidation.

General mechanism for the oxidation of alcohols and aldehydes.

Other reactions

Amide and oxime dehydration; addition of Grignard's reagents to nitriles; formation of diazonium salts; reaction of diazonium salts with CuX , KI , water, H_3PO_2 , anilines, phenols, malonic ester and acetoacetic ester; installation and removal of protecting groups (thio)acetal, THP, Bn, Me, Et, tBu, TMS, Boc, Cbz; formation of sodium, copper and silver acetylides; Arndt-Eistert's reaction; Nef's reaction; borane oxidation with peroxide to give alcohols.

Identification of organic molecules

Hinsberg's test; haloform reaction; Lucas's test; reaction with Na ; reaction with bromine water; Baeyer's reagent; Tollens's reagent; Fehling's reagent; Benedict's reagent; reaction with AgNO_3 ; reaction with FeCl_3 ; reaction with HCl ; reaction with NaOH ; reaction with Na_2CO_3 ; reaction with NaHCO_3 ; Jones' reagent; reaction with the aqueous solution of NaI and NaIO_3 ; alkaline fusion with Na followed by reaction with AgNO_3 , PbAc_2 or $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NOS}]$ and FeSO_4 .